

Ex-post design, operations and financial cost-benefit analysis of mini-grids in Ghana: What can we learn?

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ABSTRACT

This paper adopted an integrated scenario modelling approach using HOMER and financial cost-benefit analysis to investigate design accuracy, power supply challenges and financial viability of two existing mini-grids in Ghana. It also examined the economic competitiveness of grid extension relative to the mini-grid plants, and offered learnings for future improvement with broader application to other developing countries. Results of the techno-economic analysis points to over-dimensioned mini-grids which should ideally be able to supply power reliably without any issues. However, altered operating decisions led to capacity constraints which is made worse by degraded batteries. Two reasons were identified to account for the oversized plants: initial load over-estimation and the influence of grant. Cost of electricity supply was found to be considerably higher than the applied tariffs making it impossible for the recovery of even operations and maintenance cost at current levels of consumption. The analysis also show that efficiently designed tariffs, coupled with the application of existing tariff regulations can significantly improve financial performance to guarantee sustainability of the mini-grid plants. The mini-grids were observed to be cost-ineffective compared to grid extension mainly due to unsuitable siting distances. Beyond efficient tariff design, other learnings from the investigation included; (1) the need to couple in-depth surveys with modular mini-grids construction to avoid over-design, (2) the need to use diesel support to mitigate cost in systems aiming for high reliability, (3) that grants can have an expansionary effect on system dimensioning and needs to be monitored, (4) that cost-cutting measures must be an integral part of mini-grid development programmes to narrow viability gap, and that, (5) overcapacity can lead to altered operating plans with consequences.

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Introduction

About 85% of the population of Ghana are resident in communities with grid coverage but actual connection to the grid stands at 82%

Abbreviations: DG, Diesel Generator/Genset; WT, Wind turbine; CI-Actual, Initial capital inclusive systems at actual loads; CI-Design, Initial capital inclusive systems at design loads; CE-Actual, Initial capital exclusive systems at actual loads; CE-Design, Initial capital exclusive systems at design loads; GEDAP, Ghana Energy Development and Access Project; COE, Levelised cost of electricity; NPC, Net present cost; NPB, Net present benefit; BCR, Cost benefit ratio; Mft, Fixed mini-grid tariff; Gft, Fixed grid tariff; Met, Escalating mini-grid tariff; Get, Escalating grid tariff; CAGR, Cumulative annual growth rate; DoD, Depth of discharge; EDL, Economic distance limit; O&M, Operations and Maintenance; PV, Photovoltaic; VRA, Volta River Authority; CAPEX, Capital expenditure.

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(Energy Commission of Ghana, 2020). Thus, 3% of the population are not connected to the grid despite its presence. This can be attributed to low levels and seasonality of income generating activities, which leads to inability to afford connection charges and monthly electricity bills (Blimpo et al., 2020). Current deficit in access to electricity in Ghana is concentrated in rural locations as nearly 100% of urban residents have access compared to 70.5% in rural areas (Energy Commission of Ghana, 2020). Most of these unelectrified rural locations are presently bordered by the Volta Lake (which was formed by the construction of the Akosombo Dam over the Volta River in the mid-1960s), making them Islands or Lakeside communities. The remoteness of these communities and the additional geographic challenges posed by the Islands, was presumed to make the cost of reaching them with the national grid cost-ineffective (Government of Ghana, 2015). For this reason, the policy decision has been to adopt renewable energy mini-grids as the least cost option for electrifying most Island and lakeside communities in Ghana.

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From the relevant planning literature (Energy Commission of Ghana, 2019; Government of Ghana, 2015), between 300 and 400 mini-grids are intended to be deployed by 2030. These are expected to provide electricity to about 350,000 of the 2.9 million residents of the Island and Lakeside communities (Government of Ghana, 2015). With favourable solar energy resources (i.e. average global solar radiation of 3.1 kWh/m²/day - 6.5 kWh/m²/day (Quansah et al., 2017)), it is unlikely that these plans will encounter major setbacks from renewable energy resource shortfalls. What could become challenging as noted from mini-grid project assessments include the adoption of locally appropriate designs, proper capacity building, availability of spare parts and local expertise, maintenance culture, customer satisfaction, revenue adequacy, community participation and ownership, clarity of actor responsibilities, and government support (Arnaiz et al., 2018; Butchers et al., 2020; Derks & Romijn, 2019; Ferrer-Martí et al., 2012; Guerreiro & Botetzagias, 2018; Hong & Abe, 2012; Kumar et al., 2019; Odarno, 2017; Palit & Bandyopadhyay, 2017; Palit & Chaurey, 2011; Rahul Sharma & Palit, 2020; Sato et al., 2017; Sovacool, 2012; Terrapon-Pfaff, Dienst, König, Ortiz, 2014a; Ulsrud et al., 2018).

These range of technical, economic, institutional, and social factors are however observed to be poorly understood (Almashqab & Ustun, 2019; Hong & Abe, 2012; Terrapon-Pfaff, Dienst, König, Ortiz, 2014a) in spite of the growing body of evidence cited above. Some experts posit that ex-post analysis of community electrification projects are generally under-researched (López-González et al., 2018; Terrapon-Pfaff, Dienst, König, Ortiz, 2014b). Citing examples from Botswana, Zimbabwe, South Africa and Lesotho, the authors in Ref. (Azimoh et al., 2017), indicates that the narrative on mini-grid projects in Sub-Saharan Africa has been one of limited success with premature project terminations. This awareness may have incentivised the World Bank to grant fund five initial pilot projects with which the Ghanaian authorities had aimed to further the learning curve towards speedy and sustainable mini-grids electrification (Government of Ghana, 2015).

However, after five years in operation, no form of evaluative work has been conducted on the experimental projects in line with the object of gathering lessons from them as earlier noted. Generally, there is yet to be any form ex-post analysis of mini-grids in the Ghanaian context. Existing research into mini-grids in Ghana have to date been limited to pre-installation feasibility studies (Adaramola et al., 2014; Adaramola et al., 2017; AfDB, 2015; Arranz-Piera et al., 2018; Asuamah et al., 2021; Bailey et al., 2007; Ramamurthi et al., 2016) and policy assessments (Bukari et al., 2021; World Bank, 2017). One study of empirical nature focused on identifying development and climate benefits of mini-grids using surveys (Nuru et al., 2021). With a growing need to deepen understanding of the challenges faced by existing mini-grids across developing regions, this study sets out to analyse two mini-grid systems in Ghana with a focus on design, operations and returns adequacy issues. In this regard, the following specific questions will be addressed by the paper.

- Are the present power supply challenges faced by mini-grids in Ghana linked to design or operational deficiencies?
- Can the mini-grids attain financial sustainability?
- Could main grid extension be considered purely based on cost?

To accomplish these objectives, qualitative research methods, techno-economic modelling and financial cost-benefit analysis are adopted as detailed in Section 3, and in Appendices C, D, E and F in the supplementary material. The study will bridge prevailing research gaps as observed in the succeeding literature review.

In (Chmiel & Bhattacharyya, 2015), the design accuracy and performance of a 357 kW hybrid system in the Isle of Eigg Island in Scotland was assessed. The analysis points to overcapacity in the original system design compared to an alternative simulated in HOMER (Hybrid Optimization of Multiple Energy Resources). The authors however, concluded that the excess capacity aided reliability of power supply.

Although the study analysed an alternative demand option and considered the effect of grant on cost of supply, it explored limited scenarios, and does not cover aspects of financial sustainability outside the HOMER results, nor evaluate the possibility of grid extension to the Island. In (Halabi et al., 2017), the optimal design of 2.65 MW and 2.55 MW power stations were verified using HOMER in Malaysia, where their performance were contrasted against two hypothetical operating alternatives based solar PV (Photovoltaic) penetration levels. The optimal capacity selection of the simulation revealed lower capacity sizes for various components compared to the existing plants. Issues of tariff, financial sustainability and grid extension were not also considered in this study.

In (Micangeli et al., 2017), a 500 kW mini-grid system in Kenya was appraised and weighed against three alternatives based on different levels of battery storage inclusion. This study also favoured techno-economic and environmental performance comparison between the existing and the simulated alternatives and paid little attention to sizing accuracy assessment. An examination of the HOMER results however, points to under-capacity in the original system. In the context of this study (Micangeli et al., 2017), an assessment of competing electrification models and financial viability analysis would have been valuable, especially because of the insolvent nature of most African utilities (World Bank, 2016), and given that the mini-grid is utility owned, but this was not investigated. Observations of under-capacity as above, was as well seen in an assessment of a 4.1 MW plant by (Kim et al., 2017) in the Gasado Island in South Korea. The HOMER optimisation results showed an optimal system with 6.1 MW capacity. Despite the under-capacity, the authors also note the technical ability of the plant to still supply reliable power with increased use of diesel generator (DG), but at a slightly higher cost compared to the HOMER optimised system. While this study estimated economic distance limit (EDL) for grid extension, conclusions on whether grid extension was cheaper than the mini-grid could not be stated for lack of information on grid connection distance.

In (Husein et al., 2017), a further ex-post analysis is conducted on the South Korean project, where alternative cases of different renewable energy penetration levels were modelled and compared to the existing plant. The study confirmed suitability of the existing design in meeting the loads but also emphasised techno-economic, emissions and dynamic response analysis, coupling HOMER with MATLAB. (Azimoh et al., 2017), also combined HOMER and MATLAB in the study of a mini-grid project in Namibia. They focused on technical and socioeconomic impact analysis while identifying key factors for success from an operational review standpoint. It did not investigate the suitability of the installed mini-grid.

In (Veilleux et al., 2020), a failing 30 kW mini-grid system in the Koh Jik Island in Thailand is redesigned using HOMER, comparing cost implications of using lead-acid over lithium-ion batteries. Notwithstanding that this study conducted financial analysis, it did so with a private sector mindset, focusing on payback and rate of return, and without exploring the potential of alternative tariffs to improve viability. Like (Veilleux et al., 2020), (García-Vera et al., 2020) used HOMER to simulate the performance of 6 mini-grids in Columbia, and sought to optimise their performance by modifying the existing load profiles and comparing the cost of using lead-acid over lithium-ion batteries. In (Silva et al., 2010), the optimal sizing and economic assessment is rather conducted on a 5.124 kW PV-Fuel Cell pilot system at a Research Centre in Brazil with further scenarios built in HOMER to explore the possibility of extending the technology to rural communities. Additionally, (Asrari et al., 2012) assessed how economical it would have been to keep supplying electricity to a community in Iran with a 15 kW DG system by adding renewable energy generators as oppose to grid extension using HOMER. The option to still integrate the extended grid system with renewable energy resources was also explored. Others sought to validate (Zhao et al., 2014) or demonstrate (De Bosio et al., 2016) the effectiveness of implemented or developed optimisation methods on existing

mini-grids, which focus are clearly distinct from this study. The optimisation problem formulated in (Zhao et al., 2014) followed the genetic algorithm approach, while the formulation in (De Bosio et al., 2016) followed the Mixed Integration Linear Programming method.

While the review here is by no means exhaustive (and is not intended to be a systematic review), it satisfies the object of highlighting the approach and focus of previous ex-post techno-economic studies, which helps with clearly defining the contribution of the current study and provides justification for the approach adopted. The present study is distinct in many ways. First, it steps outside the HOMER results to conduct financial sustainability analysis. It explores varied tariff schemes with emphasis on balancing operational survival of the plants over their life with their potential to impose economic burden on national utilities, which is different from the profitability focus of the private sector as observed in some studies. Second, all the studies seen to be evaluating sizing accuracy have done so relying only on current loads, while this helps to assess the extent to which the capacity of an existing plant is commensurate with existing energy requirements, they overlook the important point of determining the effect of the originally estimated loads on plant sizing which is covered in this study. Third, not many studies have sought to verify the ex-ante choice of mini-grids as the most economical model of electrification when the full investment and operational costs are no longer assumed as highlighted in (Bhattacharyya & Palit, 2021). And even when they do, such studies have been unable to arrive at definite conclusions for lack of data. Finally, being the first of its kind in Ghana, this study expands the research horizon at the country level.

From the preceding review, most of the studies adopted HOMER as the preferred analytical tool. The dominant application of HOMER in techno-economic analysis of mini-grids has been widely acknowledged in literature (e.g. Refs. (Bhattacharyya & Palit, 2021; Mendes et al., 2011; Sinha & Chandel, 2014)). The tool was developed by the National Renewable Energy Laboratory for the simulation and optimisation of different mini-grid system configurations which are sorted by net present cost (NPC). HOMER incorporates a wide range of technologies for electricity, heating, and cooling generation, has a very high simulation time-step of 1 min to 1 h, allows multi-year modelling, integrates advanced storage module for in-depth battery modelling, and verify variables uncertainty through sensitivity analysis (HOMER Energy, 2022a). It is also noted to be user friendly with a rich set of exportable results tables and graphs (Sinha & Chandel, 2014). Limitations of the software includes that it remains a single objective optimisation tool and does not consider depth of discharge in the modelling of battery storage (Sinha & Chandel, 2014). Nevertheless, the consideration of state of charge and battery degradation limit in the advance storage module of the tool mitigate any risks posed by the latter in system designs.

Beyond HOMER, other tools such as the Distributed Energy Resources Customer Adoption Model (DER-CAM), the Improved Hybrid Optimization by Genetic Algorithms (iHOGA), and RETScreen *Clean Energy Management Software* (RETScreen) have also been deployed in hybrid energy systems design. According to Refs. (Berkeley Lab, 2018; Berkeley Lab, 2019; Berkeley Lab, 2022a; Berkeley Lab, 2022b; Mendes et al., 2011) DER-CAM is an economic model developed by Lawrence Berkeley National Laboratory which finds the best possible combination of energy technologies, optimal capacities, suitable placements, and operation of hybrid mini-grids, to meet both electrical and thermal loads. It also integrates a wide range of generation technologies just like HOMER. While the model originally has a single objective optimisation function of minimising total annual cost of energy supply or CO₂ emissions, it allows modifications to co-optimize other mini-grid value streams like load shifting and peak shaving. The DER-CAM algorithm also limits user intuitiveness in the optimisation process by avoiding the pre-selection of generation technologies and is found to be easily applicable. On the flipside, it assumes constant output and efficiencies for generation equipments, does not account for power quality and reliability benefits, and uses

only three representative day-types per month to model hourly energy loads which may not sufficiently represent the site-specific load profiles.

RETScreen on the other hand, is a decision support tool developed by Natural Resources Canada with a number of partners (Natural Resources Canada, 2022a). RETScreen is able to rapidly identify and assess the viability of potential energy efficiency, renewable energy and cogeneration projects (Natural Resources Canada, 2022a; Natural Resources Canada, 2022b). Advantages of the tool include that it features the most comprehensive set of power generation technologies (Connolly et al., 2010) with an extensive global climate database from more than 6700 ground stations (Natural Resources Canada, 2022c). It also has a user-friendly interface (Mendes et al., 2011). However, the software is unable to model storage systems beyond batteries, has very low simulation time step of one month and lack optimisation function. To date, the application of RETScreen to community energy systems analysis like mini-grids, as mentioned also in (Mendes et al., 2011) is almost non-existent. The exception is the work of (Salehin et al., 2016), but even here, its usage was consigned to investment analysis by further processing the HOMER results.

Also, iHOGA (details in Refs. (Universidad Zaragoza, 2021; Universidad Zaragoza, 2022a; Universidad Zaragoza, 2022b; Universidad Zaragoza, 2022c)) was developed by the University of Zaragoza, Spain, and is used to optimise hybrid energy systems involving several generation technologies. Like HOMER, the iHOGA algorithm perform simulations in time steps from 1 min to 1 h, include multiperiod simulation and optimisation, and undertake sensitivity and probability analysis. Unlike the preceding tools however, iHOGA has both mono and multi objective optimisation functions. In the default mono-objective optimisation mode, it minimises the NPC of power supply systems and maximises net present value (NPV) in generating systems. But in the multi-objective mode, equivalent CO₂ emissions or unmet load can additionally be minimised. Similarly, human development index or job creation can be maximised. Another essential feature of iHOGA is the fact that it can optimise the slope of PV panels. Nonetheless, the software has less appealing graphical representation of results and does not appear to consider parameters such as EDL and capacity shortage in the technical results, or minimum operating load in modelling fossil fuel generators.

The brief review here demonstrates the absence of a silver bullet software as every tool has its unique strengths and weaknesses. Hence, the intended objectives of a study remain the most important consideration to deciding the tool to be applied. Apart from RETScreen, it would appear based on the objectives of the current study that anyone of the remaining tools outlined here could be applied. However, HOMER was found to be nimbler and more granular in both the modelling and generated results. It allowed for the replication of the advanced parameters of components utilised on the original mini-grid plants and directly generated technical indicators such as capacity shortage, unmet load, and EDL of grid extension, which were needed to determine accuracy of design of the mini-grids. Applying the alternative models would have required estimating some of these parameters outside the software.

Organisation of the paper from start to finish, is briefly described in Fig. 1.

Study area and the mini-grid power plants

This section is broken down into four parts aimed at given readers a bird's-eye view of the communities (spatial and socio-economic background) where the mini-grids have been installed (Sections 2.1 and 2.2), of the community load profiles (Section 2.3), of the mini-grids architecture (Appendices A.1 and A.2 in supplementary material), and of the current state of operations and performance of the mini-grids (Appendix A.3 in supplementary material). The background information on the communities was mainly sourced or estimated based on Refs. (Ghana Statistical Service, 2010a; Ghana Statistical Service, 2010b; Ghana Statistical Service, 2014a; Ghana Statistical Service,

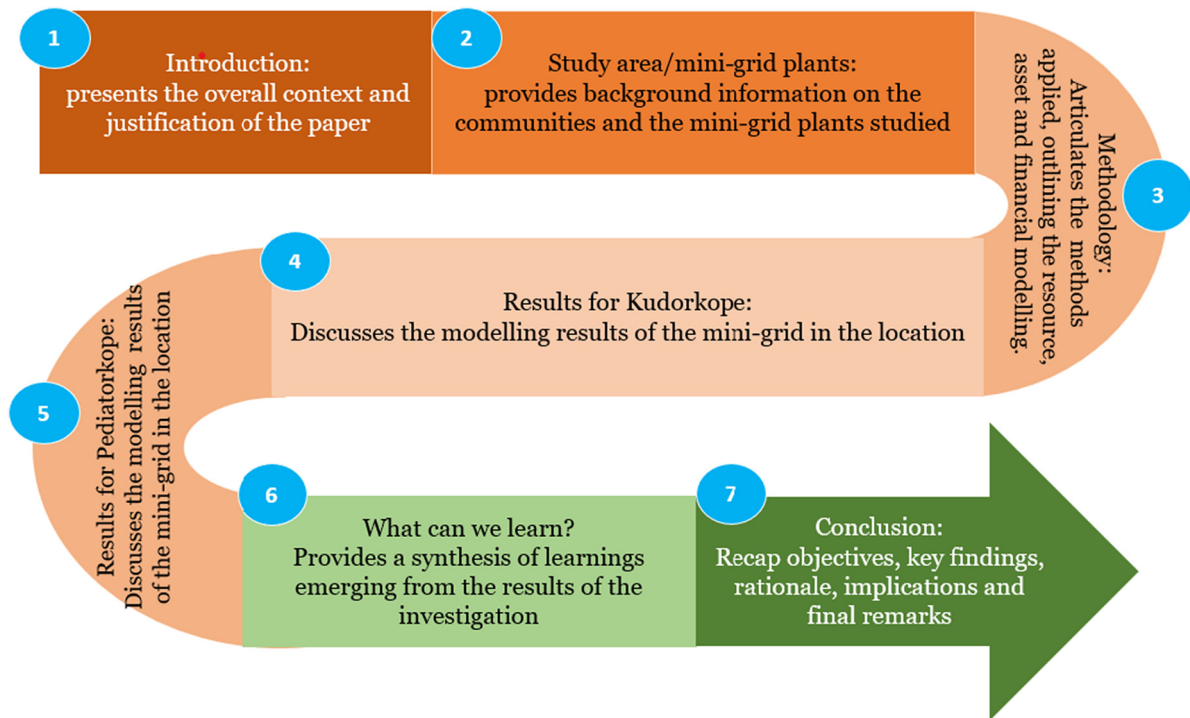


Fig. 1. Roadmap of the study.

2014b; PWC, 2012). This is supplemented with information obtained from field trips and from visualisation of the communities/the mini-grid plants on Google Earth application.

Study area 1: Kudorkope

Kudorkope is in the Krachi East District of the Oti Region. The community is listed among the twenty largest areas of the District (Ghana Statistical Service, 2014b) and is made up of six suburbs, five of which have been connected to the mini-grid plant according to the mini-grid administrator. At an elevation of about 88 m above sea level, it lies on latitude 8°03'46.76"N and longitude 0°06'55.05"E (Google Earth, 2021). The community is a clustered rural settlement (Fig. 2a) found

on the fringes of the Volta Lake and is about 14 km away from the district capital, Dambai (PWC, 2012), but based on field estimates, the community was found to be approximately 5 km away from the national grid. The pre-electrification assessment report of the projects (PWC, 2012), considered this community to be an Island but our field visits revealed otherwise. Current population of the community is projected as 1090 (Ghana Statistical Service, 2010b; Ghana Statistical Service, 2014a). With about eight people in each household (7.79) (Ghana Statistical Service, 2014a), the number of households is estimated to be approximately 140. The predominant source of income of residents is fishing (93%) (Fig. 3a), where the average monthly household income is \$986 (PWC, 2012). The average monthly pre-electrification energy expenditure was also determined as \$24.59 (PWC, 2012). By the

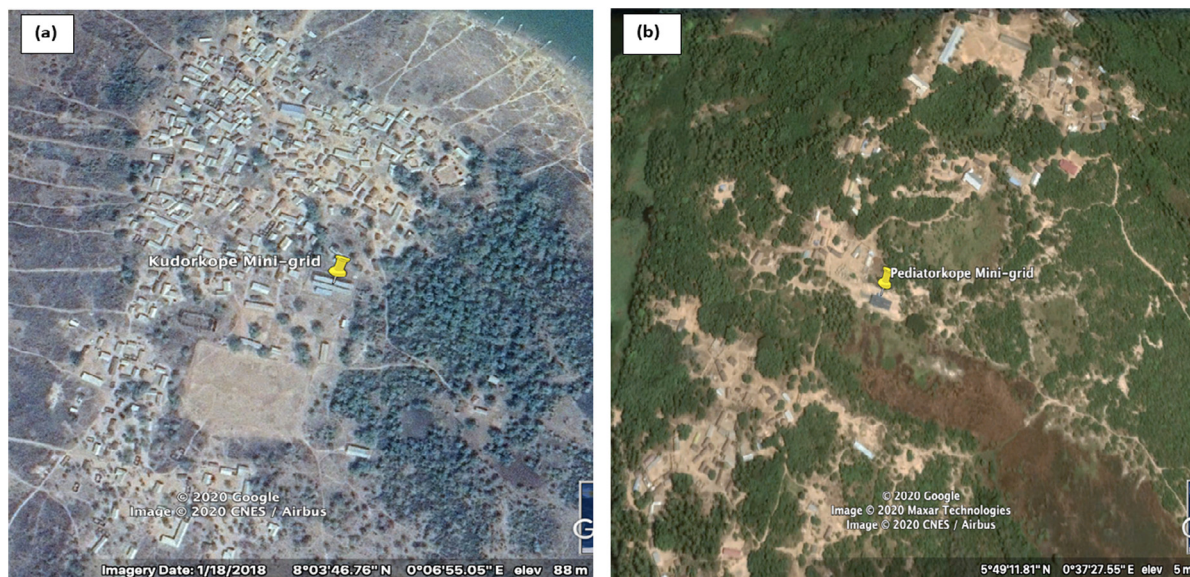


Fig. 2. Google Earth's visualisation of (a) Kudorkope and (b) Peditorkope communities/mini-grid plants.

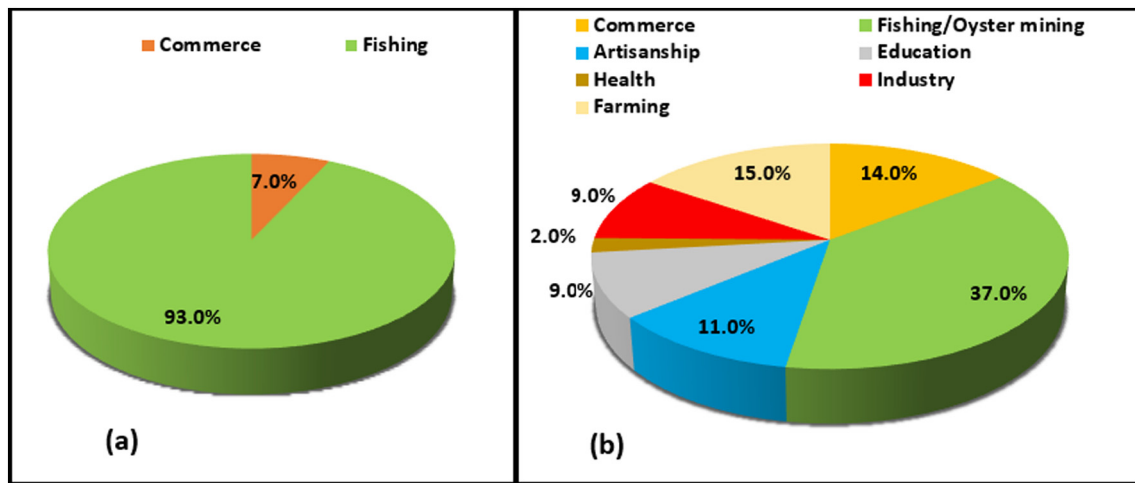


Fig. 3. Sources of income for residents in Kudorkope (a) and Peditorkope (b).

prevailing levels of income, residents of the community can be said to live above the extreme and lower middle-income poverty benchmarks of \$1.90/day and \$3.20/day respectively (World Bank, 2020).

Data Source: (PWC, 2012).

Study area 2: Peditorkope

Peditorkope is in the Ada East district of the Greater Accra Region and lies on latitude 5°49'11.81"N and longitude 0°37'27.55"E at an elevation of about 5 m above sea level (Google Earth, n.d.). It has twenty-one suburbs, seven of which are currently connected to the mini-grid according to the mini-grid administrator. The Island is about 11 km away from the district capital, Ada-foah (PWC, 2012), but from field estimates, it was found to be just about 1 km from the national grid across the Volta Lake. Discussions with officials of the Ghana Energy Development and Access Project (GEDAP) and the Ministry of Energy, revealed that electrification of the Island with submarine cables was assumed to be economically infeasible. Compared to Kudorkope, this

community is a relatively dispersed rural settlement (Fig. 2b). With a household size of 5.5, and a projected population of 6411 (Ghana Statistical Service, 2010a; Ghana Statistical Service, 2014b), the number of households is assessed to be 1107. Although fishing remains the central preoccupation of residents, the income sources of residents of this community are quite diversified (Fig. 3b), which can be explained by the peri-urban nature of the settlement. Households monthly income is about a third (\$368) that in Kudorkope (PWC, 2012). Residents of the Island are also found to live above the poverty benchmarks referenced in Section 2.1. The pre-electrification energy expense was also estimated as \$11.57 (PWC, 2012).

Existing loads

The maximum and average daily load profiles in Kudorkope (a) and Peditorkope (b) are depicted in Fig. 4. The loads, among other performance datasets were obtained from data written on SD-cards affixed to the performance monitoring units (Studer RCC-

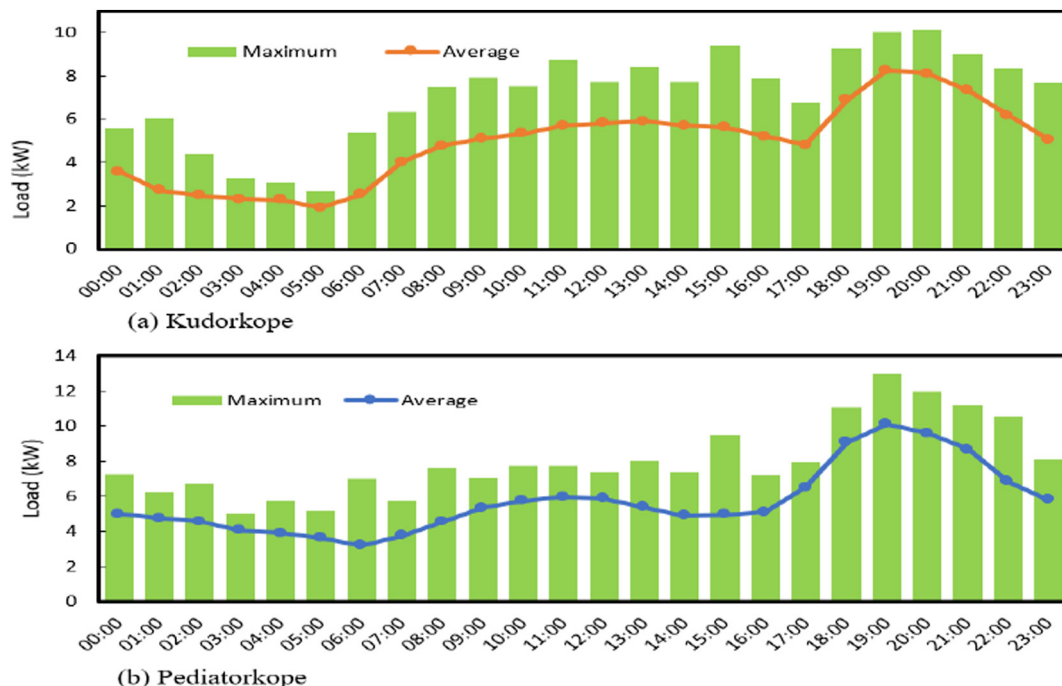


Fig. 4. Mean daily hourly profile of load in Kudorkope (a) and Peditorkope (b).

02) of the plants. Specific to loads, hourly load records of 120 days for each plant was manually extracted by visualising the dataset in the manufacturers corresponding software (Studer-Innotech, 2020). Priority was given to days with limited disruptions to data records. The process yielded an average total electrical load of 117.65 kWh/day in Kudorkope and 137.29 kWh/day in Peditorkope. As observed from Fig. 4, peak load in both locations occur between 6 pm and 10 pm with base loads occurring at late night. Following the prescription in Ref. (Williams et al., 2017), day-to-day and timestep variability parameters were respectively estimated as 11.65% and 30.35% in Kudorkope and 14.2% and 27.8% in Peditorkope. Peak power at both sites were estimated to be 16.63 kW in Kudorkope and 19.10 kW in Peditorkope giving the corresponding load factors as 29.0% and 30.0%. With about 175 customers connected to the mini-grid in Kudorkope and 176 in Peditorkope, per capita electricity consumption at both sites are estimated to be 0.67 kWh/day in Kudorkope and

0.78 kWh/day in Peditorkope. Table A2 in Appendix B (in supplementary material) provides detailed information on current consumer profiles, energy subscriptions and the tariffs levied on customers under the mini-grid systems. Electricity consumption is predominantly residential at both locations with an average residential electricity tariff of \$0.115/kWh which is lower than the average paid by end-users on the main grid (\$0.21/kWh) (Energy Commission of Ghana, 2020). Tariffs levied to consumers under the mini-grids have not seen any upward adjustment since 2016.

Methodology

An integrated scenario modelling approach involving five main contiguous steps, as labelled in Fig. 5, was employed as the analytical framework for this paper. The framework is adapted from (Bhattacharyya, 2015) to meet the needs of the current study. At the pre-HOMER

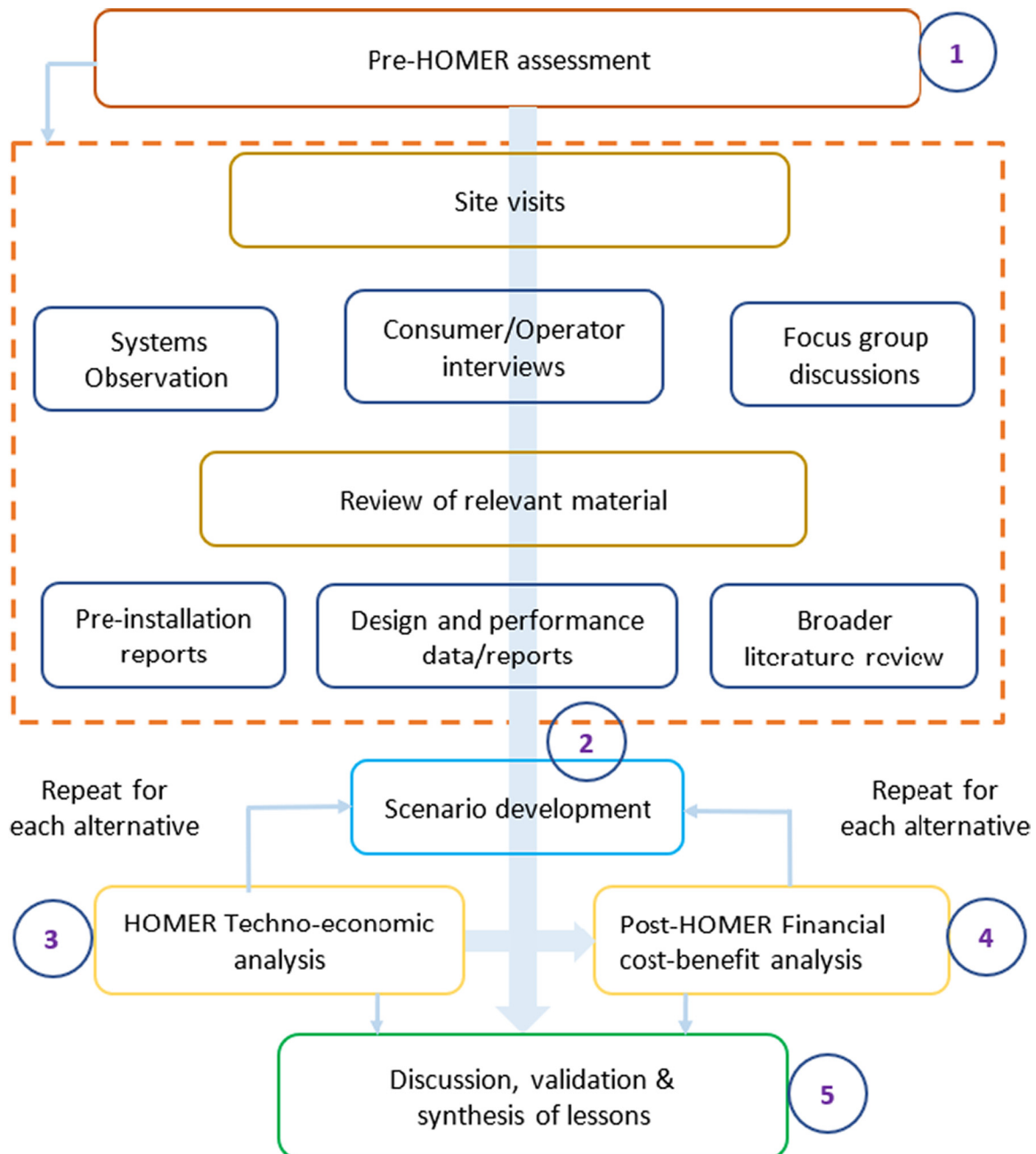


Fig. 5. Analytical framework of the study. (Source: Authors' construct).

assessment stage, site visits were conducted to observe the engineering set up of the mini-grid plants, and to get insights into the operations, management and performance of the systems through interviews with end-users and the local administrators, in addition to focus group discussions with the local energy committees. In the second leg of the pre-HOMER assessment step, project specific data and reports were retrieved from GEDAP, VRA, the local administrators and the monitoring units installed on the plants. These, together with broader literature reviews provided further illumination on the mini-grid plants which helped to shape the study objectives and the modelling approach. It should be emphasised that Stage 1 was more iterative than linear.

Insights from the first stage, especially the discovery of unreliable power supply, and the fact that the gensets were not being operated to support reliable power supply, informed the development of scenarios in the second stage. First, to determine if the systems were optimally installed and are capable of reliably meeting the current loads. And second, to ascertain if the vulnerabilities discovered can be linked to the decision of not operating the gensets. The specific scenarios developed were the “Installed System Scenario (Scenario 1)”, which simulated the existing architecture, the “Operating System Scenario (Scenario 2)”, which simulated the current operating architecture and the “Alternative Scenario (Scenario 3)”, which examined the accuracy of configurations of the preceding scenarios. These were executed under two load conditions: the existing and the design loads. The design loads as reported in the project design documents are 187.81 kWh/day for Kudorkope and 157.41 kWh/day for Peditorkope (Trama TecnoAmbiental, Arthur Energy Services, GTS Engineering Services, 2015a; Trama TecnoAmbiental, Arthur Energy Services, GTS Engineering Services, 2015b). In the Installed and Operating Scenarios, the asset capacities are constraint to limits on the original plants. In the Alternative Scenario however, the simulation tool is freely allowed to decide component capacities without such limitations. The Alternative Scenarios determines the optimality of design of the preceding scenarios by reconfiguring the available resources within larger search spaces.

HOMER Pro version 3.14.4 was used to accomplish the scenarios in the HOMER techno-economic analysis stage. The effect of grant on cost of supply and sizing optimality were also considered in simulating the Installed and Operating Scenarios. Thus, four cases were explored in those scenarios. These are the initial capital inclusive systems at actual loads (CI-Actual) and design loads (CI-Design), and the initial capital exclusive systems at actual loads (CE-Actual) and design loads (CE-Design). Modelling of the Alternative systems were however limited to capital inclusion as true optimisation requires the reflection of complete economic cost to be sufficient. Results of the most optimal systems which HOMER selects based on the lowest NPC were adopted for analysis. The primary metrics used to determine if the plants were accurately designed and are capable of reliably meeting the existing and design loads, were relative capacity size selections, useful energy served, levelised cost of energy (COE), unmet load and capacity shortage.

At the Post-HOMER financial analysis stage, cash flow analysis were conducted in Microsoft excel following the financial cost-benefit analysis approach adopted in (Bhattacharyya, 2015; Palit et al., 2011), to scrutinise financial viability of the mini-grid plants. By the approach, a system is seen to be financially sustainable if the present value of costs is less than the present value of benefits, or at least, the present value of costs must be equal to the present value of benefits. In other words, the ratio of benefits relative to costs (BCR) should be greater than or equal to 1. The analysis explored the effect of four different tariff options on financial performance. These comprises the fixed mini-grid tariff (Mft), the fixed grid tariff (Gft), the escalating mini-grid tariff (Met) and the escalating grid tariff (Get).

On one hand, Mft examines the effect of the current average fixed tariffs (\$0.15/kWh) charged to customers of the mini-grid plants, while Gft explores what financial performance would be if the tariffs in 2016 were fixed at par with that charged to residential grid

consumers (\$0.21/kWh). On the other hand, Met assesses financial outcomes under circumstances where Mft is subjected to periodic adjustments (usually quarterly) within the multi-year tariff framework of the economic regulator of utilities in Ghana, the Public Utilities Regulatory Commission (PURC). Get, similarly reflect these adjustments to grid parity tariff (Gft) to ascertain the influence of the electricity pricing regime on financial stability of the mini-grids. Further information on electricity tariff setting by PURC can be found in (PURC, 2018).

The exact execution of modelling of components of the mini-grids and energy resources, as well as the economic and financial modelling, are respectively explained in Appendices C, D and E in the supplementary material, while Appendix F explains the modelling procedure to determining if centralised grid extension would have been more economical. In the final stage, the results are discussed with key findings validated through further conversations with the local administrators and system managers at VRA, where key lessons were then outlined.

Results and analysis of the Kudorkope mini-grid

This section discusses the technical and financial results of scenarios under the Kudorkope plant. For both this, and the Peditorkope plant, the results are presented for the three different scenarios mentioned in Section 3: the “Installed”, the “Operating” and the “Alternative” Scenarios. The Installed Scenario describes the results of replicating the originally installed mini-grids architecture in the simulation tool while the Operating Scenario describes the results of simulating the operating architecture in both locations. Thus, the DG units which are not being operated, are excluded from the Operating Scenario simulations. Results of the Alternative Scenario on the other hand, gives an account of what the ideal or optimised system architecture of both mini-grids should be for the installed and operating system configurations, without regard to any capacity thresholds.

In each scenario, the results are presented in three parts, the technical, financial and the grid versus mini-grids comparative results. Analysis of the technical results presents four cases in each scenario, covering the actual and design loads, where for each load, the results capture the inclusion of initial capital investment or expenditure (CAPEX) in the simulation, referred to as capital inclusive systems (CI-Actual and CI-Design), and the exclusion of CAPEX in the simulation, referred to as capital exclusive systems (CE-Actual and CE-Design). Corresponding to the four technical cases in each scenario, the financial analysis also present results for the four tariff options explained in Section 3, for which further details are captured in Appendix E2. It should however be emphasised that the technical cases examined under the Alternative Scenario were limited to the capital inclusive systems. Appendix F sufficiently explains how the comparative economic valuation of mini-grids and the central grid was executed, except to add that, that analysis relates only to the Installed Scenarios where investments into the original mini-grids were fully reflected.

Analysis of the Installed Scenario

The analysis presented here aims to establish if the installed plant can meet existing loads as well as preliminarily determine if the mini-grid was oversized. The financial viability and grid competitive assessments for the scenario follow thereafter. The dispatch strategies deployed by HOMER in this scenario was found to be the load following strategy in the CE-Actual, CE-Design and CI-Actual cases, but in the CI-Design case, this changed to cycle charging strategy. By way of definition, the load following strategy refers to the situation where anytime a genset is dispatched, it just produces enough power to meet primary loads, leaving the renewable power sources to meet other energy needs like charging the storage bank or serving deferrable loads. In contrast, the cycle charging strategy requires that if a genset is dispatched to serve the primary loads, it would be operated at full output power, where the surplus electrical energy will go to meet other energy

Table 1
Simulation results of the installed mini-grid in Kudorkope.

Parameter	CI-Actual	CI-Design	CE-Actual	CE-Design
Solar PV	30 kW	40 kW	54 kW	54 kW
Battery	24 Units	24 Units	48 Units	48 Units
Inverter	16 kW	16 kW	16 kW	16 kW
Generator	26.4 kW	26.4 kW	26.4 kW	26.4 kW
Energy served	42,942 kWh/year	68,551 kWh/year	42,942 kWh/year	68,551 kWh/year
Unmet Load	0%	0%	0%	0%
Capacity Shortage	0%	0%	0%	0%
NPC	\$491,736	\$622,402	\$117,426	\$289,229
COE	\$0.836/kWh	\$0.663/kWh	\$0.302/kWh	\$0.308/kWh

Table 2
Financial results of the installed mini-grid in Kudorkope.

Parameter	CI-Actual	CI-Design	CE-Actual	CE-Design
BCR @Mft	0.14	0.17	0.38	0.37
BCR @Gft	0.25	0.32	0.70	0.68
BCR @Met	0.40	0.51	1.11	1.09
BCR @Get	0.73	0.93	2.04	1.99

needs as previously mentioned. Additional information on these strategies can be found in (HOMER Energy, 2022b).

Simulation results of the Installed Scenario

The results for this scenario can be found in Table 1. Without accounting for capital grant financing, a little above one-half the capacity of PV (0.56), one-half battery and one-third inverter capacities of the original plant are deemed to be adequate to meet the existing loads. In the design load case, only PV capacity changes to around three-quarters of the capacity installed. The only commonality between the selections here and that which exist in the field is the DG component size of 26.4 kW which can also be considered to be excessive for the current loads. The rule of thumb is for DGs to be sized to meet peak loads (Adaramola et al., 2014; Louie, 2018), which in this case is only 16.63

kW. Nonetheless, the realised peak load of 26.55 kW for the design loads suggest that the oversized DG may have been influenced by over-estimation of the initial loads. Such attribution cannot however be made regarding the totality of the overbuilt plant, since the selected architecture even in the design load case is lower than installed in the field.

Taking initial grants to account, the configurations change to assume the precise capacity limits of PV, battery, and DG on the existing plant in both the actual and design load cases, with inverter capacity remaining one-third as observed earlier. Accordingly, these latter configurations suggest that the generally oversized nature of the plant may be driven by grant. The oversized nature of the inverters is especially pronounced since the recommendation is for margins to remain between 20 and 40% above system continuous loads (Louie, 2018; Studer-Innotec, 2018), yet, the inverters installed at the site are observed to be 81% higher than the peak load of 26.55 kW in the design load cases. The expected energy requirements are observed to be fully met in all cases, at 0% capacity shortages and unmet loads. COEs of the various systems simulated are as well captured in Table 1.

Financial results of the Installed Scenario

As observed from Table 2, static application of the mini-grid tariff or even grid parity tariff, with or without grant support, yielded negative net present benefits (NPB) in all cases. Positive NPB become possible under tariff reviews but only with grant support (Fig. 6). An examination of the detailed financial results shows that benefits at the fixed mini-grid tariff, are incapable of covering O&M cost in the actual load cases, with or without grant support. In the design load cases however, 24% of fuel cost in the CI-Design and 52% in the CE-Design become recoverable over O&M. Similarly applying the grid tariff, tends to fully recover fuel cost beyond O&M under both load conditions in the cost reflective systems. And in the grant supported cases, 38% of replacement cost can be recovered. Thus, regardless of capital grant considerations, financial sustainability appears unfeasible with static tariffs, and even granting replacement costs as additional subventions would still make the recovery of O&M and fuel costs difficult under the mini-grid tariff.

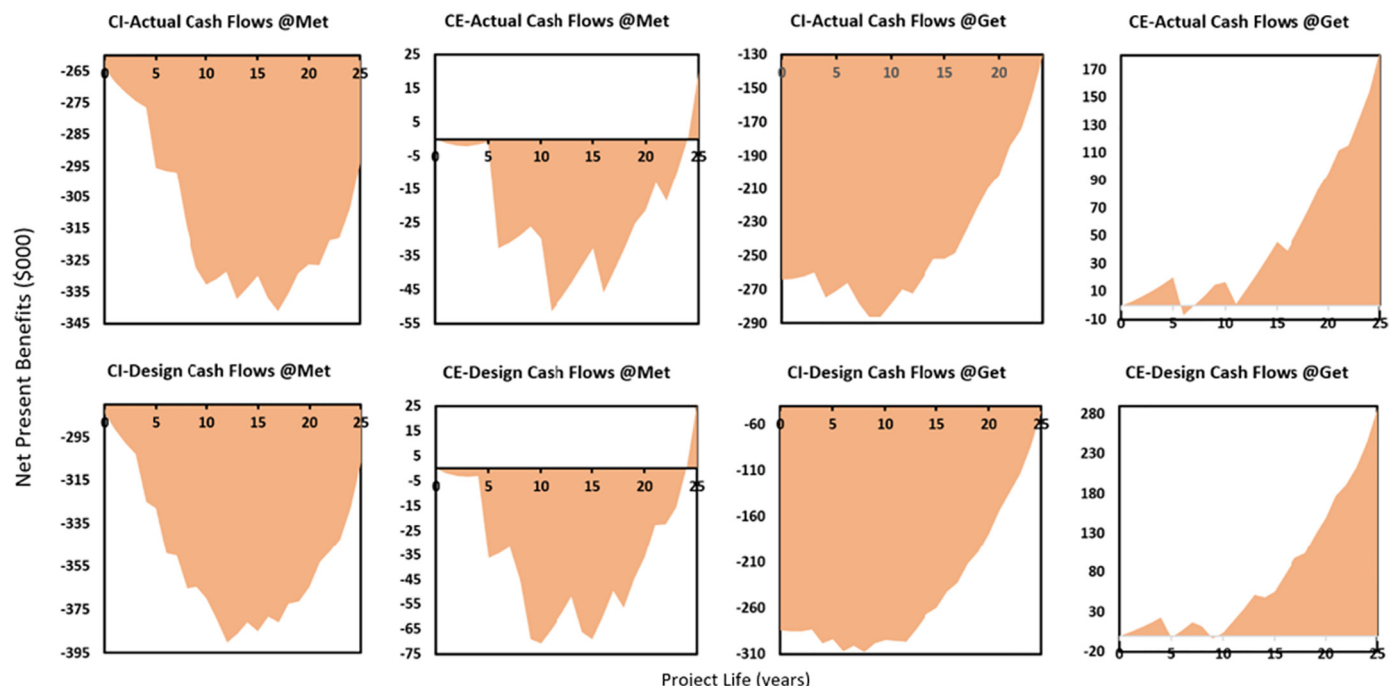


Fig. 6. Net benefits of the Installed Scenario under tariff review in Kudorkope.

While combining tariff reviews with grant support led to positive NPB for both tariff cases (Fig. 6), the cases with tariff reviews alone still falls short of recovering depreciated asset cost in the CI-Actual (37%) and CI-Design (21%) cases under the mini-grid tariff. With the grid tariff however, 46% of initial capital is recovered in the CI-Actual and up to 82% in the CI-Design. Thus, absolute financial sustainability appears a nonstarter. But tariff reviews may be more effective at reaching financial sustainability compared to capital grant support (Table 2). The annual revenue requirements under the current mini-grid tariff are respectively determined as \$30,961, \$37,565, \$8030 and \$13,230 for the CI-Actual, CI-Design, CE-Actual and CE-Design.

Grid Extension versus mini-grid

Regarding the economic prudence of installing the standalone mini-grid as opposed to grid base electrification, the EDL for grid extension was estimated as depicted in Fig. 7. The breakeven grid extension distance which is the distance that makes the NPC of extending the grid equal to that of a standalone system (HOMER Energy, 2020) was determined as 9.45 km. With a grid point connection distance of only 5 km, it can be argued that extending the national grid to the community would have been the cheapest mode of electrification. The spurious classification of this community as an Island must have been the main misleading factor, which questions the level of diligence of pre-electrification planning processes. Generally, the results imply that for a mini-grid to be cheaper than grid extension at present cost, the grid connection distance must be roughly 10 km from the community to be electrified.

Analysis of the operating scenario

The analysis presented here aims to establish if the operating plant can reliably meet the existing loads. As the DG unit is barely used, the key point of the technical analysis here is to evaluate the performance of the plant without the genset. Thus, the technical results are initially presented, with the financial viability assessment following thereafter. HOMER deployed cycle charging as the dispatch strategy for all cases (CE-Actual, CE-Design, CI-Actual and CI-Design) in this scenario.

Simulation results of the operating scenario

No feasible solution could be generated initially by HOMER (Fig. 8) at the given load and resource inputs for both load cases, regardless of whether grant support was taken into account. Possible reasons were given for the occurrence (Fig. 8). Only three of the assigned reasons are relevant for the situation simulated here as emission reduction

limits were not applied. Capacity shortage was deliberately set to zero for complete reliability assessment considerations. Being a pure PV system, 100% renewable fraction was automatically imposed. The third reason relates to inadequate generation capacity, which really, is the subject of interest here: whether the pure PV system as presently operated, can reliably meet the existing and design loads. The outcome in Fig. 8 point to difficulties in that regard. But to better understand the situation, capacity shortage sensitivity range of 0.5%–5% was implemented as recommended in (Givler & Lilienthal, 2005). Once the recommended range failed to deliver any feasible solution, a wider capacity shortage constraint range (5%–35%) was applied. From an economic point of view, HOMER then selected 30 kW PV and 48 batteries together with 8 kW inverter for the actual load, which resulted in unmet load ratio of 17.1%, caused by capacity shortage of 21.7%. But since the selected architecture was yet to exhaust the capacity limits of the existing plant, an accurate assessment of how reliable the actual loads were being met could not be made. To generate that understanding, PV capacity was adjusted to 54 kW. The results in Table 3 demonstrate that indeed, the operating plant is incapable of effectively meeting the existing loads, given the unmet load ratio of 5.6%, which are occasioned by capacity shortfall of around 8.5%.

Also, the simulation results generated a battery end-of-life 5.7 years at the current loads which suggest that the batteries may now be technically “dead” for reasonable performance. Hence, the power supply shortages will be more severe than observed here. Once the assessment showed that the current loads could not be met by the operating plant, the expectation of elevated power supply shortages became obvious for the design loads (Table 3). The outcome of the design loads is however not surprising as the design objective of the plant for that load was for PV to be complemented by a DG unit in the delivery of power. The revealed levels of power supply, match the unmet loads and capacity shortages established. Also, the COEs for various cases can be found in Table 3.

Financial results of the operating scenario

Financial results of this scenario (Table 4) also showed that even with grant support, financial viability cannot be attained except by combining tariff reviews (Fig. 9). Akin to findings in Section 4.1.2, the accrued benefits under the fixed mini-grid tariff, will be insufficient to cover O&M cost in the actual load conditions, with or without capital inclusion, but becomes adequate to service O&M in the design loads cases. Static grid tariffs lead to partial recoveries of replacement costs in the actual (43%) and design (68%) load cases beyond O&M. In the grant unsupported cases where adjusting tariffs fail to attain financial sustainability, varying

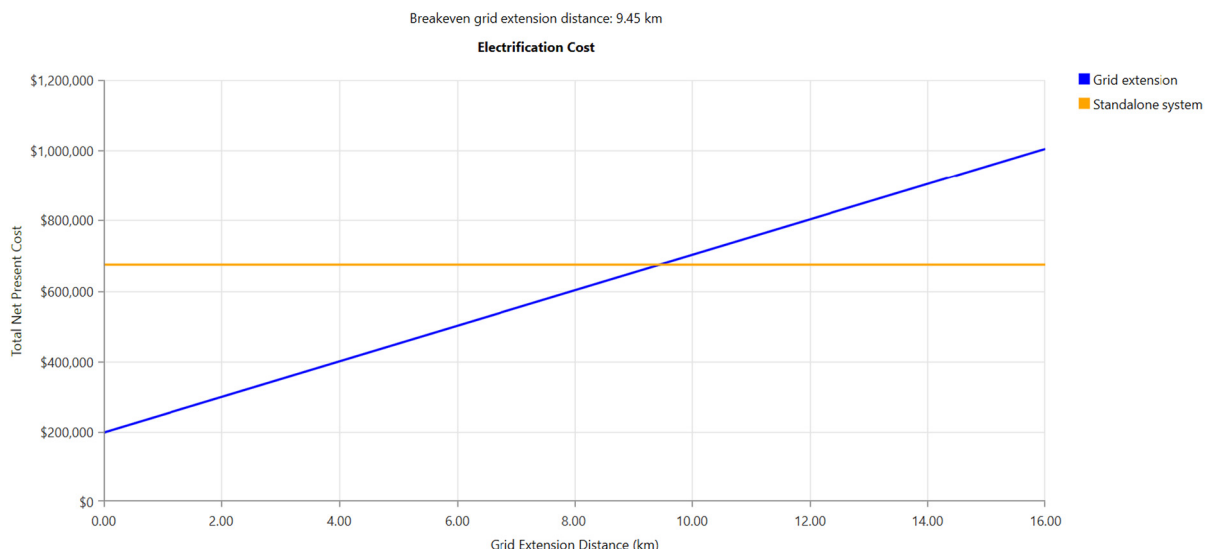


Fig. 7. Grid extension valuation of Kudorkope mini-grid plant.

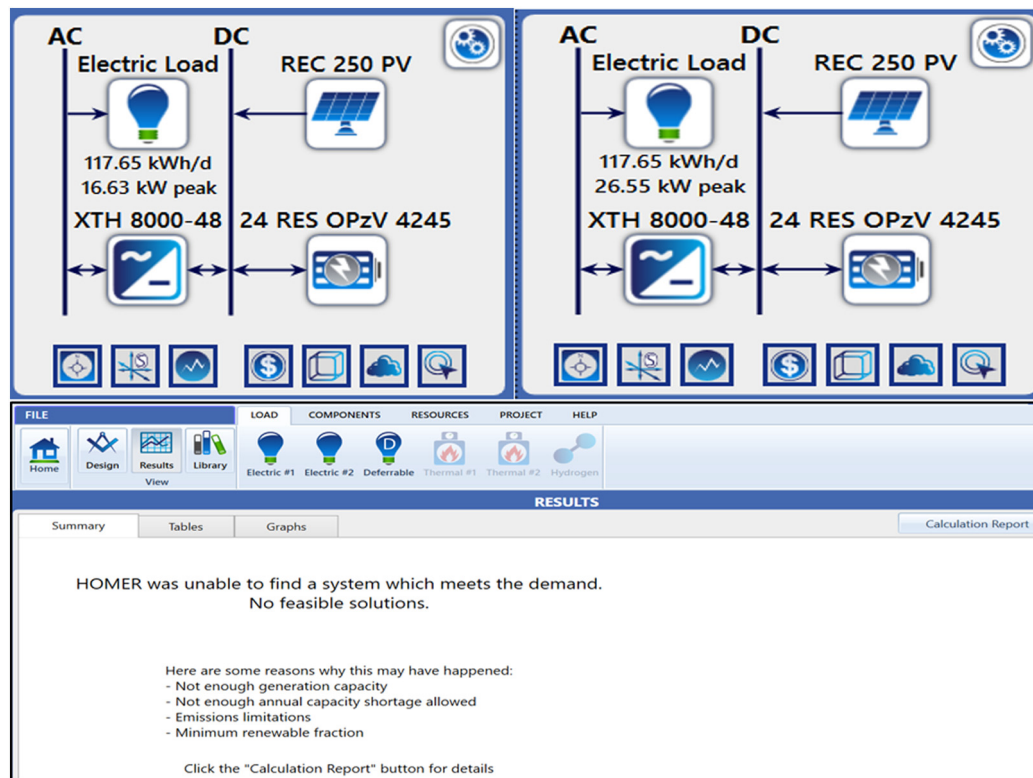


Fig. 8. Infeasible solution of the operating mini-grid system in Kudorkope.

degrees of initial capital become recoverable. The covering ratio range from 6% for the CI-Actual to 19% for the CI-Design, with the escalating mini-grid tariff. These ratios improve to 58% for the CI-Actual and 84% for CI-Design when the escalating grid parity tariff is implemented. Annual revenue deficits under the prevailing mini-grid tariff are respectively estimated as \$28,550, \$29,141, \$7016, and \$7079 for the CI-Actual, CI-Design, CE-Actual and CE-Design cases. The conclusions are similar to Section 4.1.2 except that capacity shortage appears to slightly improve cost recovery as well as lower the viability gap.

Analysis of the alternative scenario

The analysis presented here aims to determine what the optimal systems configuration and capacity limits of the mini-grids should have been for the scenarios simulated. The financial viability analysis follows for each scenario. For the Alternative Installed Scenario, HOMER deployed load following as the dispatch strategy for the CI-Actual case and cycle charging for the CI-Design case relative to the technical results. In the case of the Alternative Operated Scenario, cycle charging dispatch strategy is adopted for both cases.

Table 3
Simulation results of the operating mini-grid in Kudorkope.

Parameter	CI-Actual	CI-Design	CE-Actual	CE-Design
Solar PV	54 kW	54 kW	54 kW	54 kW
Battery Capacity	48 Units	48 Units	48 Units	48 Units
Inverter	8 kW	16 kW	8 kW	16 kW
Energy served	40,555 kWh/year	51,669 kWh/year	40,555 kWh/year	51,669 kWh/year
Unmet Load	5.56%	24.6%	5.56%	24.6%
Capacity Shortage	8.45%	29.4%	8.45%	29.4%
NPC	\$455,190	\$480,473	\$160,188	\$178,527
COE	\$0.819/kWh	\$0.679/kWh	\$0.288/kWh	\$0.252/kWh

Optimisation results of the Alternative Installed Scenario

Compared to Section 4.1.1, optimising the installed system (Table 5), slightly reconfigure the mini-grid architecture in the actual load case, where PV and battery capacities increases slightly to reduce diesel use from 10.3 L/day to 9.13 L/day while the inverter capacity is further reduced to 11 kW. Just about a single bank battery is determined to be sufficient in the actual load case as was the case in the constraint optimisation. With respect to the design load case, PV and battery capacities are further reduced compared to the constraint. The minimised architecture under both load cases yet meet the expected energy requirements reliably in the actual and design load cases at 0% unmet loads and capacity shortages. Thus, the underlining fact of oversizing of the Kudorkope mini-grid was confirmed. Comparing COEs here to the constraint optimisation results shows a decline due to system optimisation to \$0.818/kWh in the actual load case, and \$0.660/kWh in the design load case.

Financial results of the Alternative Installed Scenario

As observed in Table 6, NPB remain negative from all standpoints, even for the cases involving tariff reviews (Fig. 10). Expense recoveries mimic that in the initial simulations under static tariff applications. Benefits are unable to pay for O&M in the actual load case but amortises 24% of fuel cost above O&M with the mini-grid tariff. With grid tariff, O&M and fuel costs are fully recovered under both loads cases. Reviewed tariffs recover varying degrees of replacement and initial capital costs. Here, replacement cost recoveries are respectively 82% and 77% in the

Table 4
Financial results of the operating mini-grid in Kudorkope.

Parameter	CI-Actual	CI-Design	CE-Actual	CE-Design
BCR @Mft	0.16	0.17	0.44	0.46
BCR @Gft	0.30	0.31	0.81	0.83
BCR @Met	0.48	0.50	1.30	1.33
BCR @Get	0.87	0.90	2.37	2.43

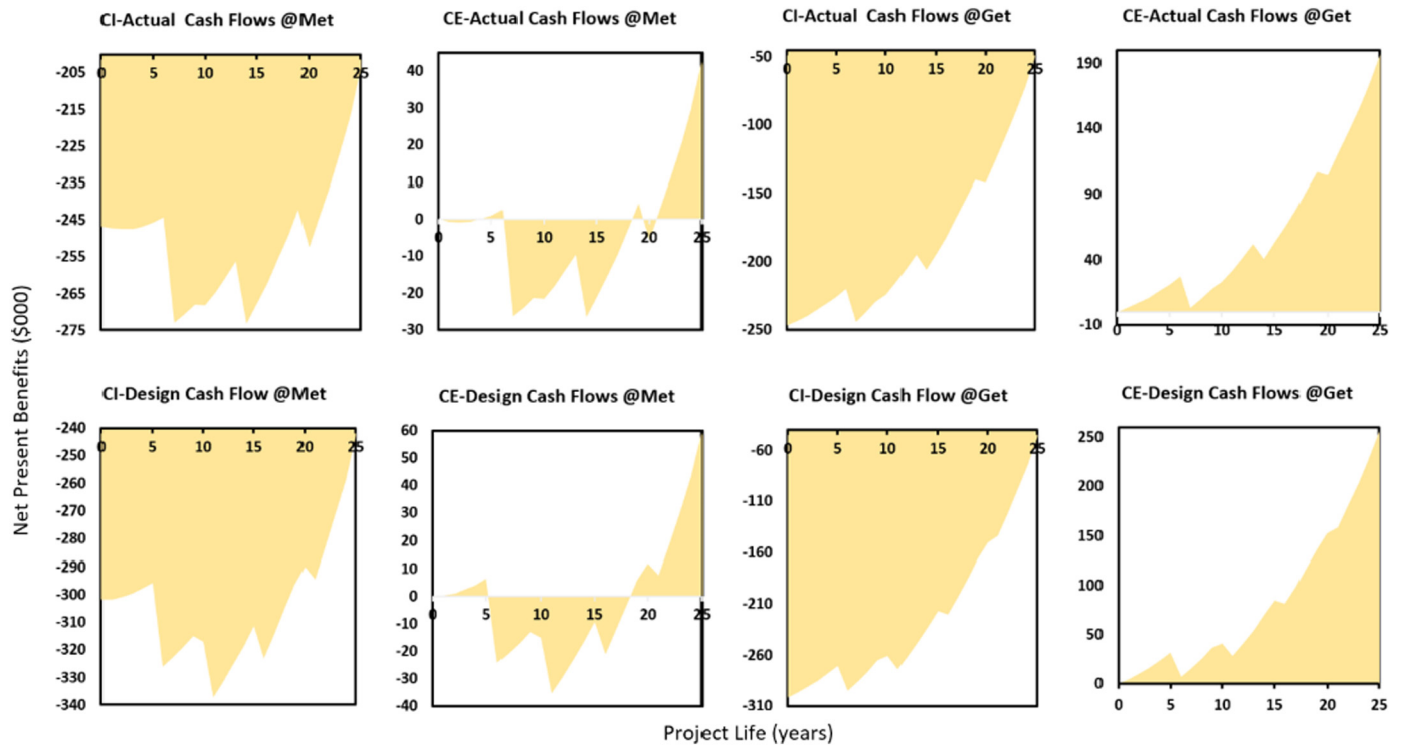


Fig. 9. Net benefits of the Operating Scenario in Kudorkope under tariff review.

actual and design load cases, when the escalating mini-grid tariff is applied. Similarly, an adjusting grid tariff recovers 54% and 82% of initial capital corresponding to the actual and design load cases. Compared to the constraint simulation, annual revenue requirement under current mini-grid tariff marginally drops to \$30,188 and \$37,360 for the CI-Actual and CI-Design respectively. The results here conclusively establish impracticability of reaching financial viability, but the capacity for tariff reviews to anchor long term survival of mini-grids has also been demonstrated which is interesting as capital grants will not always be available.

Optimisation results of the Alternative Operating Scenario

The operating architecture was indicated to be inadequate relative to meeting both the existing and design loads from the initial simulation. By optimising that architecture (Table 5), storage capacity expands by 1.5 times in the actual load case compared to the field installations. In the design load case, storage expansion goes up to 2.75 times while PV capacity is increased by one-half. However, only one-half of the existing inverter capacity is indicated to be required even at the design loads,

Table 5
Optimisation results of the alternative installed and operating plant in Kudorkope.

Parameter	Alternative Installed Scenario		Alternative Operating Scenario	
	CI-Actual	CI-Design	CI-Actual	CI-Design
Solar PV	34 kW	37 kW	50 kW	80 kW
Battery Capacity	25 Units	23 Units	120 Units	180 Units
Inverter	11 kW	17 kW	16 kW	24 kW
Generator	26.4 kW	26.4 kW	0 kW	0 kW
Energy served	42,942 kWh/year	68,551 kWh/year	42,941 kWh/year	68,517 kWh/year
Unmet Load	0%	0%	0.002%	0.05%
Capacity Shortage	0%	0%	0.02%	0.09%
NPC	\$481,394	\$619,803	\$560,146	\$759,953
COE	\$0.818/kWh	\$0.660/kWh	\$0.952/kWh	\$0.810/kWh

while that in the actual load case is one-third. The results here underline constraints of the existing plant to operate reliably as a pure PV system at current loads. It also signals the architecture needed to ensure uninterrupted power supply as the previously observed capacity shortages and unmet loads have become practically eliminated (Table 5). The expanded system sizes result in higher COEs of \$0.952/kWh and \$0.810/kWh for the actual and design load cases respectively.

Financial results of the Alternative Operating Scenario

As expected, the optimised operating plant results in worsened financial performance (Table 6) compared to the constraint operating system. This can be explained by the additional investments needed by the operating plant to supply electricity at high reliability. Operational cost coverage is also relatively worsened. Financial returns from the fixed mini-grid tariff, cannot cover O&M cost in the CI-Actual but barely does so in the CI-Design case. The grid parity tariff leads to lower recovery of replacement costs of 36% in the actual and 60% in the design load cases. Like the fixed tariff cases, tariff reviews fail to generate positive NPV (Fig. 10). And the improved cost coverage is limited to covering replacement costs in both load cases with the mini-grid tariff. With grid tariff reviews, benefits can pay for lower level of capital cost; 44% of in the actual load case and 61% in the design load case. Surely, the annual revenue requirement at the existing tariff also worsens to \$35,942 and \$47,619 in the actual and design load cases

Table 6
Financial results of the alternative installed and operating systems in Kudorkope.

Parameter	Alternative Installed Scenario		Alternative Operating Scenario	
	CI-Actual	CI-Design	CI-Actual	CI-Design
BCR @ Mft	0.14	0.17	0.12	0.14
BCR @ Gft	0.26	0.32	0.22	0.26
BCR @ Met	0.41	0.51	0.35	0.42
BCR @ Get	0.75	0.93	0.64	0.76

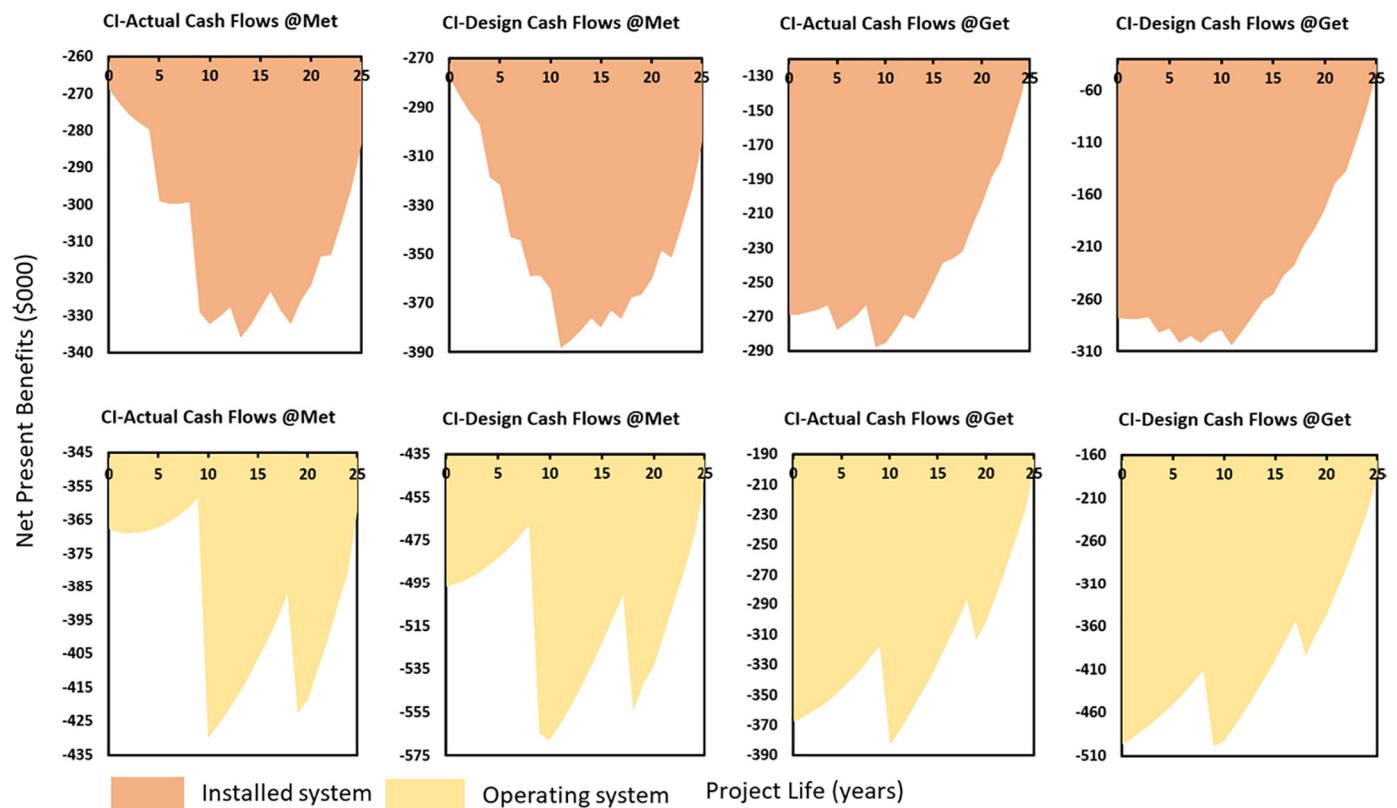


Fig. 10. Net benefits of the alternative installed and operating plants under tariff review in Kudorkope.

respectively. Thus, aiming for high reliability with the PV only system here worsens financial performance.

Results and Analysis of the Pediatorkope Mini-grid

This section discusses the technical and financial results of scenarios under the Pediatorkope plant. The presentation and character of the scenarios follow the same pattern as explained in Section 4.

Analysis of the Installed Scenario

The aim of this section corresponds to that in Section 4.1 except this pertains to the Pediatorkope plant. In terms of the dispatch strategies, HOMER deployed the load following strategy for all cases in this scenario.

Simulation results of the Installed Scenario

The sizing results in this scenario (Table 7), present a configuration lower than the plant at site for the CI-Actual case, where PV (0.23) and battery capacities reduces by one-quarter and inverter by two-thirds. For the CE-Actual and the design load cases however, the capacity limits of PV, battery and DG as exist in the field have been replicated. In all cases, the inverter capacity remains lower by two-thirds compared the original plant, but the existing DG capacity is repeated. It appears that the DG capacity selected here could also be lower for both loads given the peaks loads of 19.01 kW for actual and 21.90 kW for the design. In all instances, the selected configurations meet the expected energy requirements at 0% unmet loads and capacity shortages. The outcome in the CI-Actual case signals that the existing plant architecture is larger than required for the existing loads. However, having the architecture of the CI-Design case mirror the original system configuration except for inverters suggest that the existing over capacity was influenced by load estimation error. But also, the CE-Actual results

demonstrate the expansionary effect of grant on system dimensioning as previously observed. Again, the COEs of the varying cases can be found in Table 7.

Financial results of the Installed Scenario

Just as we saw in the corresponding scenario in Section 4.1.2, the financial results here (Table 8) point to the difficulty of attaining financial sustainability except by combining tariff reviews with grant support (Fig. 11). Benefits at the fixed mini-grid tariff, are not adequate to cover O&M cost in the actual load cases but cover 37% of fuel cost above O&M in the design load cases. Thus, even the award of replacement cost as additional subventions will have fuel cost unrecovered with or without capital grant support. Applying the grid parity tariff leads to a lag in asset replacement cost recoveries by 75% and 59% in the actual and design load cases respectively. As tariff reviews with grant support present positive NPB, the capital inclusive systems also become able to completely cover replacement costs in the actual and

Table 7

Simulation results of the installed system scenario in Pediatorkope.

Parameter	CI-Actual	CI-Design	CE-Actual	CE-Design
Solar PV	30 kW	39 kW	39 kW	39 kW
Wind Turbine	2 WT	2 WT	2 WT	2 WT
Battery	36 Units	48 Units	48 Units	48 Units
Capacity				
Inverter	16 kW	16 kW	16 kW	16 kW
Generator	26.4 kW	26.4 kW	26.4 kW	26.4 kW
Energy Served	50,104 kWh/year	57,455 kWh/year	50,104 kWh/year	57,455 kWh/year
Unmet Load	0%	0%	0%	0%
Capacity	0%	0%	0%	0%
Shortage				
NPC	\$504,958	\$542,629	\$194,921	\$231,214
COE	\$0.736/kWh	\$0.689/kWh	\$0.284/kWh	\$0.294/kWh

Table 8
Financial results of the installed mini-grid system in Pediatorkope.

Parameter	CI-Actual	CI-Design	CE-Actual	CE-Design
BCR @Mft	0.16	0.17	40	0.39
BCR @Gft	0.29	0.30	0.74	0.71
BCR @Met	0.46	0.49	1.18	1.14
BCR @Get	0.83	0.89	2.16	2.09

design load cases even with the mini-grid tariff. Recoveries at a reviewed grid tariff, now tend to include 69% and 78% of capital cost in the actual and design load cases respectively. As previously observed, tariff reviews may avail sufficient benefits than capital grants when compared separately (Table 8). Thus, while sustainability is still difficult, tariff reviews may assure operational longevity of the plant given the extent of capital cost recovered. Annual revenue requirements at the present mini-grid tariff are similar compared to Section 4.1.2 as follows; \$31,115, \$32,979, \$8468, and \$10,284 for the CI-Actual, CI-Design, CE-Actual and CE-Design cases respectively.

Main-grid versus mini-grid analysis in Pediatorkope

Results of the estimated EDL for grid extension relative to the Pediatorkope mini-grid plant is shown in Fig. 12. The breakeven grid extension distance is found to be 1.21 km. The results here imply that in Islands where submarine cables will be required, grid connection point must be beyond 1.2 km for standalone mini-grids to be more economical than the main-grid. While this is such a very short distance and many Islands may be farther than observed, in the specific case evaluated, the distance was found to be sufficient to make grid extension more cost-efficient. The grid point connection distance across the river was assessed to be around 1 km. Since the cost estimates utilised here are specific to cabling under the seabed, cost is likely to be lower in the circumstances of Ghana where rivers are involved. The possibility also exists for centralised grid extension to this community as there exist a lake Island on which poles can connect from the riverbanks. Thus, the

fundamental question of due diligence in pre-electrification assessments is again being raised.

Analysis of the Operating Scenario

The aim of this section corresponds to that in Section 4.2 except this pertains to the Pediatorkope plant. The dispatch strategy deployed by HOMER here was found to be cycle charging for all cases simulated.

Simulation results of the Operating Scenario

Like the operating scenario in Kudorkope, no feasible solution was initially generated. Same reasons as explained in Section 4.2.1 have been provided (Fig. 13). Consequently, capacity shortage adjustments led to the selection of 39 KW PV, 1 WT, 8 kW inverters and 48 batteries for the actual load cases, which resulted in unmet loads of 19% caused by capacity shortage of 25%. Again, this configuration is lower compared to the operating plant at site, and so to enable conclusions on whether the existing plant can meet the existing loads adjustments had to made where the WT was adjusted to 2 which led to the results in Table 9. Herein, the energy delivered is 45,204 kWh/year, at unmet loads of 10%, motivated by capacity shortage 14%. Thus, the inability of the operating plant to reliably meet the existing loads was established. The simulation results also presented battery end-of-life of 5.1 years for the actual loads which implies that the power shortages experienced in the field may be more severe than observed here. In the case of the design loads (Table 9), unmet loads and capacity shortages were determined to be 16% and 21% respectively. In all cases here one-sixth (1 unit) of the existing inverter capacity was indicated to be required. But to form a three-phase grid with the kind of inverters utilised, minimum 2 units (16 kW) are required (Studer-Innotec, 2018). COEs for the cases can be seen in Table 9.

Financial results of the Operating Scenario

Financial analysis here (Table 10) also point to negative NPB for all cases with fixed application of both the prevailing mini-grid and grid parity tariffs. As observed in the matching scenario for the Kudorkope

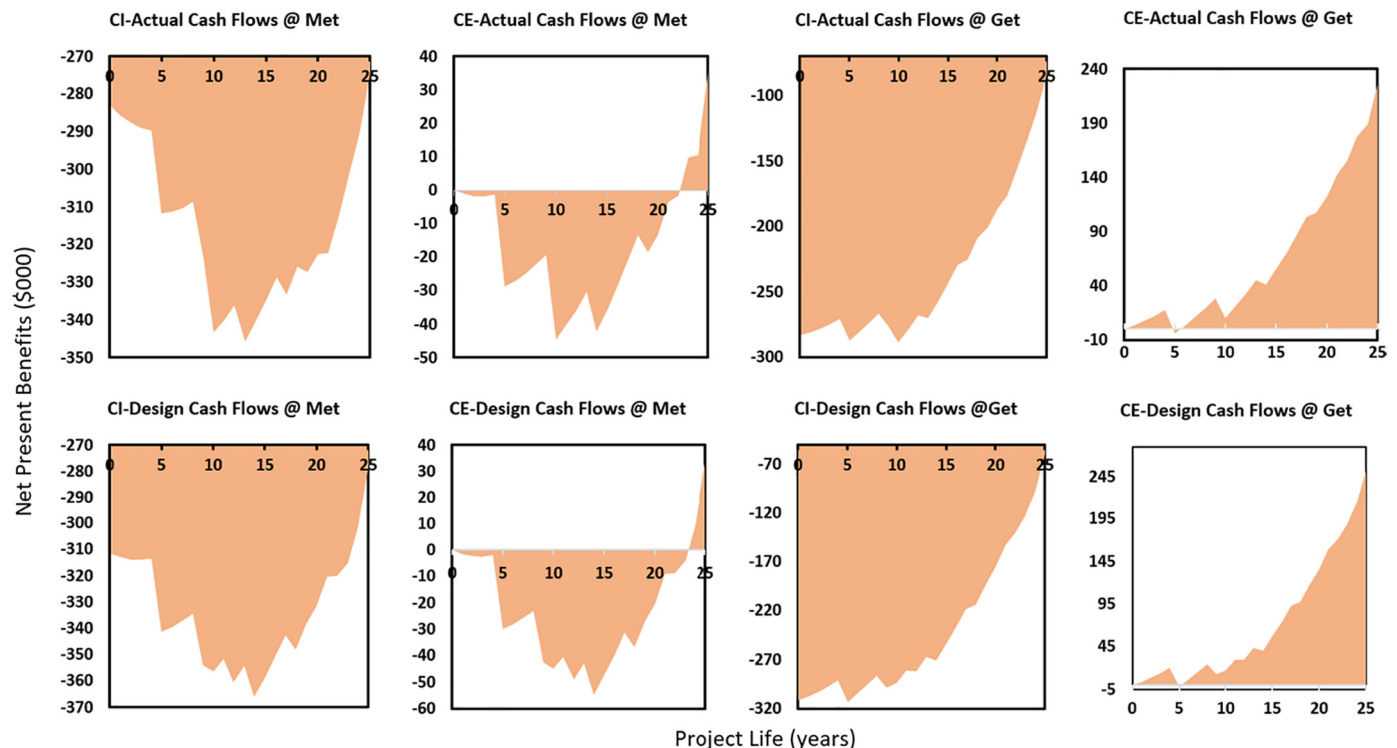


Fig. 11. Net benefits of the installed mini-grid system in Pediatorkope under tariff review.

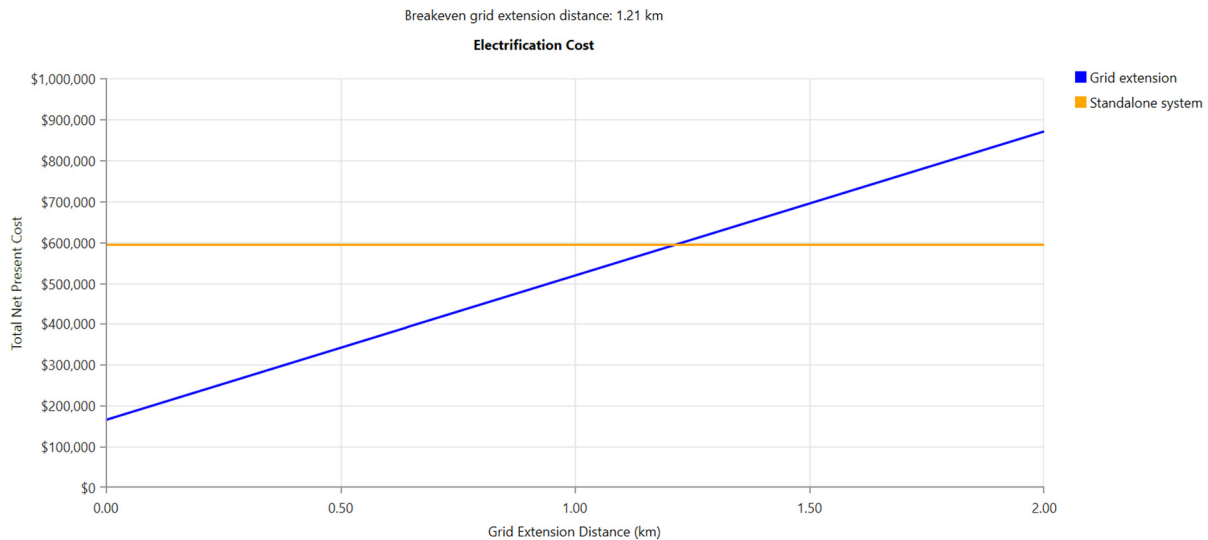


Fig. 12. Breakeven grid extension distance of the installed mini-grid in Peditorkope.

plant, financial returns are unable to cover O&M cost in the actual load cases. In the design loads however, benefits become adequate just to meet O&M cost. At the grid parity tariff, replacement costs can be recovered at varying degrees in the actual load (64%) and design load (67%) cases. When the tariffs come under regulatory reviews, positive NPB are observed only in the grant supported cases (Fig. 14). In the grant excluded cases, capital expense recovery of around 18% and 19% become observable with the adjusting mini-grid tariff and 82% and 88% with the adjusting grid tariff, for the actual and design loads respectively. Annual revenue requirement under current application of the mini-grid is estimated as \$26,083, \$25,860 \$6419 and \$6210 in the CI-Actual, CI-Design, CE-Actual and CE-Design cases respectively. Again, the conclusions are similar to Section 5.1.2 although capacity shortage appears to slightly improve cost recovery as well as lower the revenue gap.

Analysis of the Alternative Scenario

Again, the aim of this section corresponds to that in Section 5.3 except this pertains to the Peditorkope plant. With respect to the dispatch strategy deployed by HOMER, the simulation results presented the load following as the dispatch strategy for all cases in the Alternative Installed Scenario and cycle charging for all cases in the Alternative Operated Scenario.

Optimisation results of the Alternative Installed Scenario

By fully optimising the Peditorkope plant, configuration of the installed mini-grid radically changed (Table 11). Wind turbine capacity increased exponentially, curtailing PV and storage. In the CI-Actual case, WT capacity increased by 2.5 times, battery decreased by one-third and

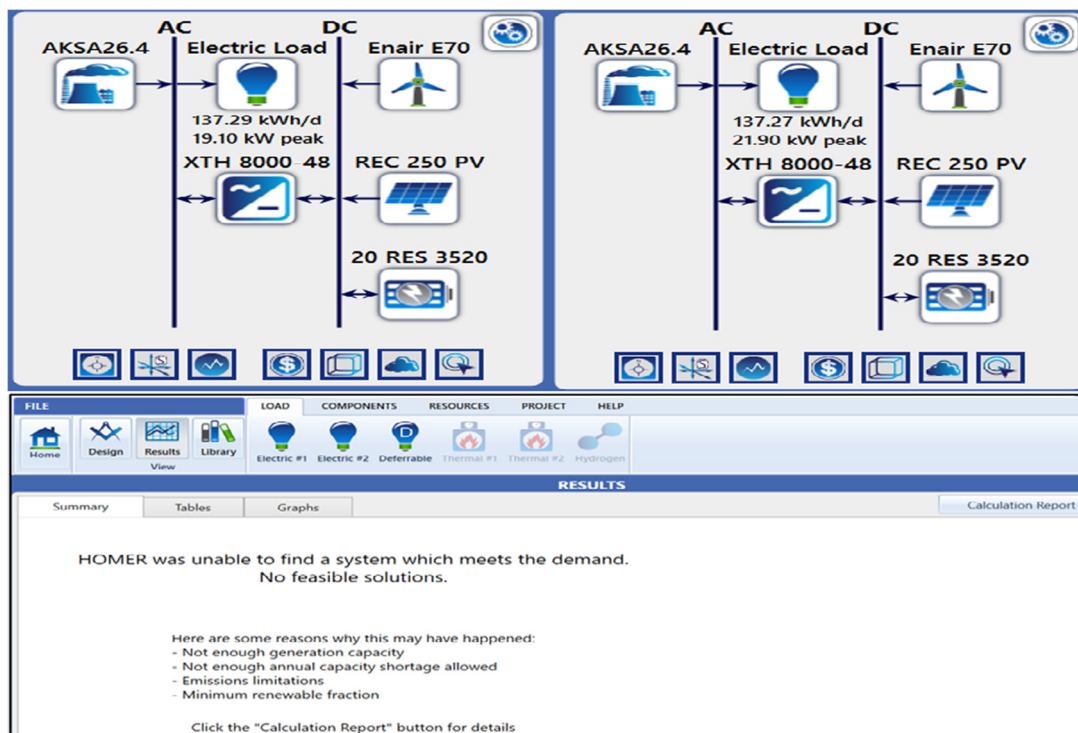


Fig. 13. Infeasibility of the operating mini-grid in Peditorkope at 0% capacity shortage.

Table 9
Simulation results of the operating mini-grid in Pediatorkope.

Parameter	CI-Actual	CI-Design	CE-Actual	CE-Design
Solar PV	39 kW	39 kW	39 kW	39 kW
Wind Turbine	2 WT	2WT	2 WT	2 WT
Battery	48 Units	48 Units	48 Units	48 Units
Inverter	8 kW	8 kW	8 kW	8 kW
Energy Served	45,204 kWh/year	48,519 kWh/year	45,204 kWh/year	48,519 kWh/year
Unmet Load	9.8%	15.6%	9.8%	15.6%
Capacity Shortage	13.7%	21.0%	13.7%	21.0%
NPC	\$428,499	\$430,854	\$159,278	\$161,633
COE	\$0.692/kWh	\$0.648/kWh	\$0.257 /kWh	\$0.243/kWh

Table 10
Financial results of the operating mini-grid in Pediatorkope.

Parameter	CI-Actual	CI-Design	CE-Actual	CE-Design
BCR @Mft	0.17	0.18	0.45	0.47
BCR @Gft	0.30	0.32	0.82	0.86
BCR @Met	0.49	0.52	1.31	1.38
BCR @Get	0.89	0.95	2.39	2.53

just about one-half the existing PV capacity is indicated to be required. In the CI-Design case, PV, battery and inverter capacity remain lower than exist in the installed plant. WT capacity increased further to 3.5 times, but the decrease in battery and PV tapers to one-fifth and a little below one-half respectively. Again, the DG unit remain common to the HOMER optimised configuration and the original plant in both load cases but only one-third of existing inverter capacity has been selected. The complete reorientation of the mini-grid architecture, which led to lower cost of supply compared to the corresponding scenario (Section 5.1.1) suggest that the mini-grid in Pediatorkope is not only oversized, but also not efficiently configured given the site resources. The optimal selections meet the expected energy supplies in the actual and design load cases at 0% unmet loads and capacity

Table 11
Optimisation results of the Alternative Installed and Operating mini-grid in Pediatorkope.

Parameter	Alternative Installed Scenario		Alternative Operating Scenario	
	CI-Actual	CI-Design	CI-Actual	CI-Design
Solar PV	20 kW	22 kW	35 kW	40 kW
Wind Turbine	7 WT	9 WT	5 WT	7 WT
Battery	33 Units	39 Units	90 Units	100 Units
Inverter	16 kW	16 kW	19 kW	22 kW
Generator	26.4 kW	26.4 kW	0 kW	0 kW
Energy Served	50,104 kWh/year	57,455 kWh/year	50,070 kWh/year	57,418 kWh/year
Unmet Load	0%	0%	0.06%	0.06%
Capacity Shortage	0%	0%	0.09%	0.09%
NPC	\$479,255	\$512,430	\$511,735	\$559,304
COE	\$0.698/kWh	\$0.651/kWh	\$0.746/kWh	\$0.711/kWh

shortages. The determined COEs are respectively 0.698/kWh and \$0.651/kWh for the actual and design load cases.

Financial results of the Alternative Installed Scenario

Like the results in Section 5.1.2, NPB continue to be negative from all angles (Table 12), even under tariff reviews (Fig. 15). Thus, although the cost reductions achieved through optimising the plant improved expense recovery as seen here, they are not enough to overturn the deficits. Benefits remain limited to covering O&M in the actual load case and some portion of fuel cost (35%) in the design load case, with the mini-grid tariff. Asset replacement cost falls short by 30% in the actual and 20% in the design load cases when the grid parity tariff is applied. If tariffs were to be adjusted in accordance with regulation, O&M, fuel, and replacement costs will become fully recovered in all cases in addition to 20% of capital cost in the actual and 26% design load case when the mini-grid tariff is applied. With grid parity tariff, capital expense is nearly entirely recovered; 86% in the actual load case and 97% in the design load case. Again, the results here reassert that if tariffs follow regulation cost recovery can be significantly improved. The annual revenue requirement at the current mini-grid tariff is lowered compared to

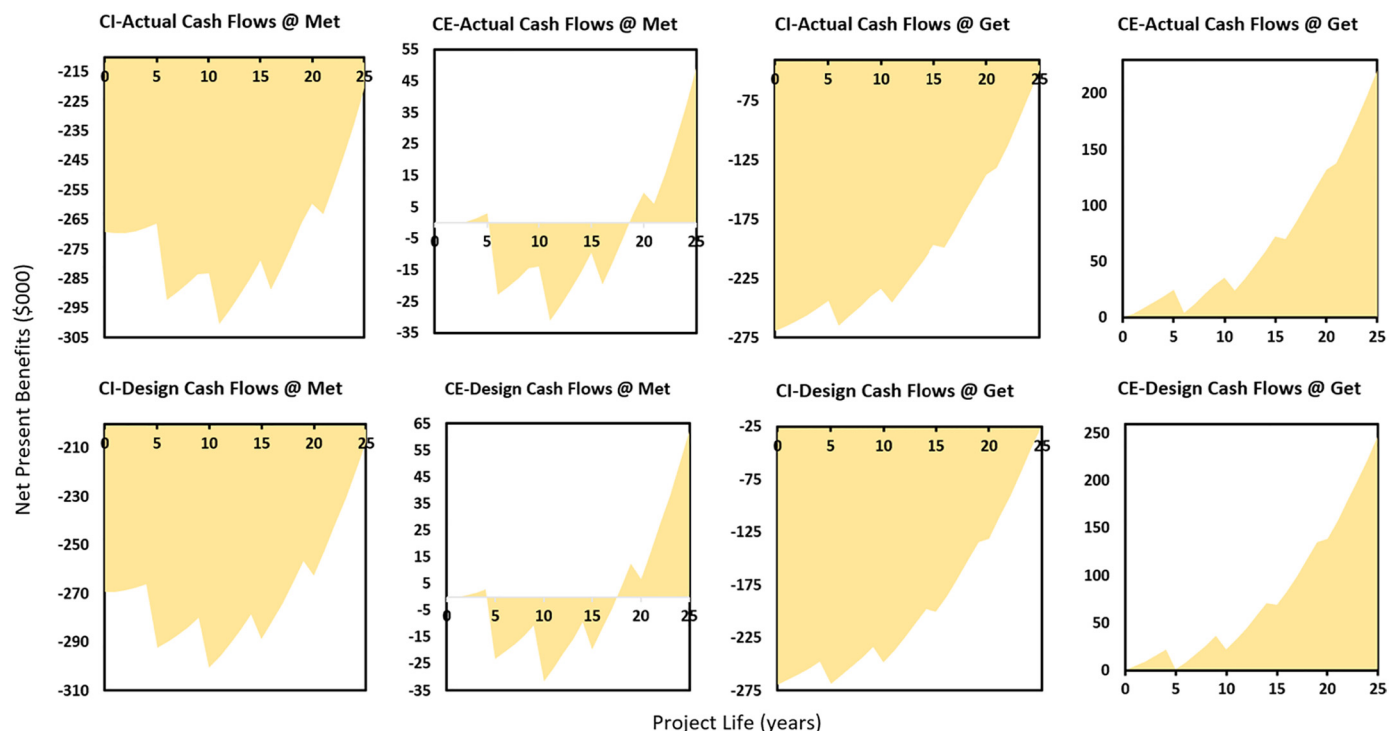
**Fig. 14.** Net benefits of the operating mini-grid plant under tariff review in Pediatorkope.

Table 12
Financial results of the Alternative Installed and Operating Scenarios.

Parameter	Alternative Installed Scenario		Alternative Operating Scenario	
	CI-Actual	CI-Design	CI-Actual	CI-Design
BCR @Mft	0.16	0.18	0.15	0.16
BCR @Gft	0.30	0.32	0.28	0.30
BCR @Met	0.48	0.52	0.45	0.47
BCR @Get	0.88	0.94	0.82	0.86

Section 5.1.2 to \$29,211 and \$30,796 for the actual and design load cases respectively.

Optimisation results of the Alternative Operating System Scenario

Optimising the operating plant (Table 11), also substantially changed its configuration, with steep increases not just in WT but also in storage capacity, compared to that which exist at the site. Wind turbine and battery capacities respectively increased by 1.5 and 2.5 times, and 0.88 and 1.1 times in the actual and design load cases. PV capacity is lower in the actual (35 kW) and but similar to what exist in the design (40 kW) load case. The optimised inverter capacities indicate that less than half the existing are required in the actual (0.4) and design (0.46) load cases respectively. For a pure renewable system, the surge in storage capacity is completely expected. The results affirm the findings of capacity constraints in Section 5.2.1, and signals what the rightful system architecture should be if the plant were intended to be operated as a pure PV-WT plant. The optimal systems selected reliably meet the community loads with inconsequential levels of unmet loads and capacity shortages of 0.06% and 0.09% in the actual and design load cases respectively. COE increases compared to Section 5.2.1 to \$0.746/kWh in the actual and \$0.711/kWh design load case.

Financial results of the Alternative Operating Scenario

From Table 12, optimising the operating plant expectedly worsened NPB compared to the constraint operating scenario. This is owed to the

different capabilities of the plants. Negative NPB are realised for the actual and design loads cases from the tariff perspectives evaluated, including the option of tariff reviews (Fig. 15). At the existing mini-grid tariff, benefits marginally fall short of covering O&M cost in the CI-Actual but does so CI-Design case. The fixed grid parity tariff results in the recovery O&M cost in both load cases in addition to replacement costs recoveries in the actual (55%) and design (63%) load cases. With mini-grid tariff review, capital expense of 10% and 14% in the actual and design load cases respectively, become recoverable over O&M and replacement costs. On the other hand, review of grid tariff enables 69% of capital expense recovery in the actual load case and 76% in the design load case. Annual revenue requirement under the present mini-grid tariff is estimated as \$31,594 in the CI-Actual and \$34,221 in the CI-Design. Comparatively, financial performance of Alternative Installed systems looks better than the Alternative Operating (Tables 12 & Fig. 15).

Discussion: what can we learn?

As seen from the constraint and unconstraint optimisation results of the installed systems, both mini-grids were found to be oversized mainly due to load over-estimation and the influence of grant. In Kudorkope where such oversizing is relatively high, the excess of load estimates compared to current loads yet remain at about 37.4%. By contrasting cost between the existing and the optimised systems architecture, the associated cost of such errors amounts to \$96,014 (27%) of capital cost, \$28,422 (23%) of replacement cost and \$8113 (10%) of O&M cost. The overall potential cost avoidance in net present cost terms comes to \$83,675 over the project life. Indeed, the accuracy of survey predicted loads as basis for initial system designs has generally been questioned with varied reported impact as observed in Refs. (Blodgett et al., 2017; Hartvigsson & Ahlgren, 2018; Louie & Dauenhauer, 2016) which stresses the need to be meticulous in their conduct to improve effectiveness. Thus, the first learning from the assessment here is that in-depth surveys (i.e. surveys thoroughly designed and methodically executed, following the recommendations of (GIZ, 2016) for example) can be coupled with modular mini-grids construction to avoid overdesign.

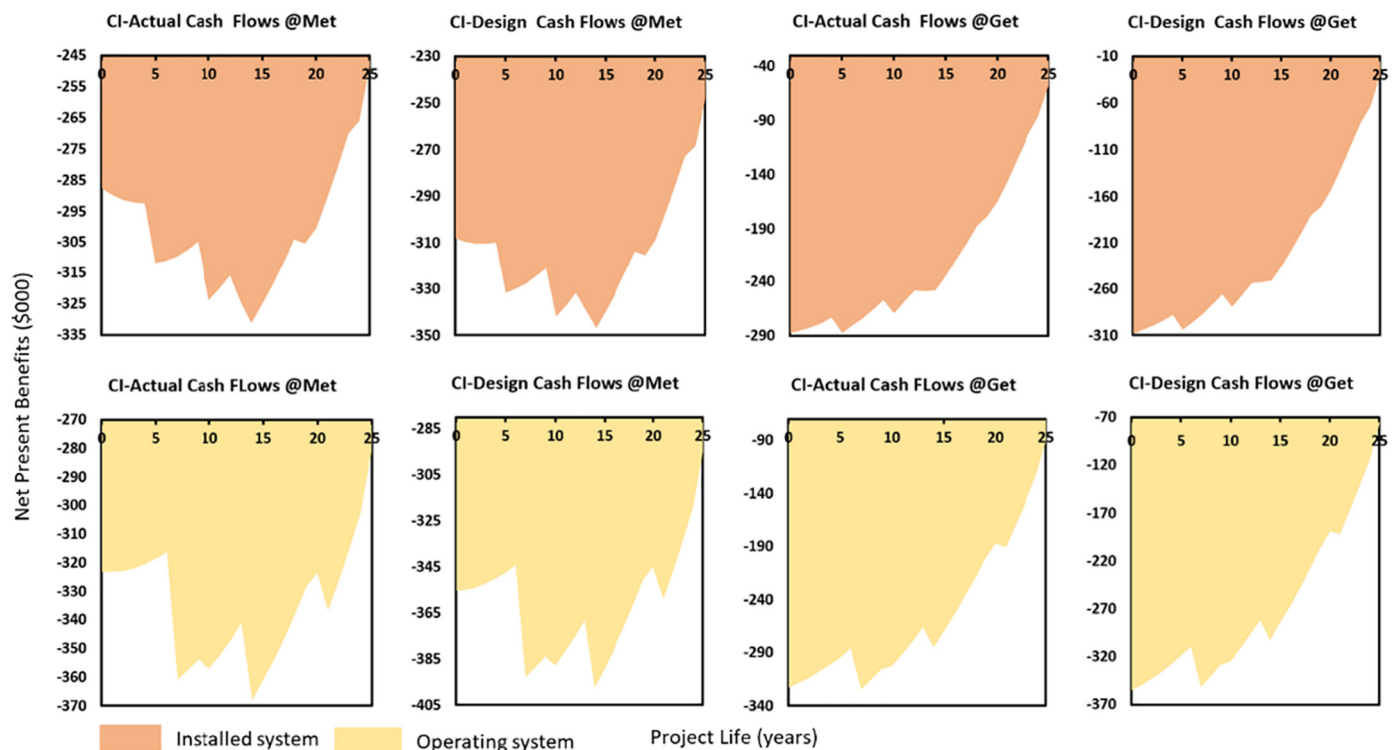


Fig. 15. Cumulative cash flows of the Alternative Installed and Operating Scenarios under tariff review.

But in places like Ghana with pre-existing mini-grid load profiles now, extrapolating load inputs from them for future mini-grids development might be most useful.

Related to the above is the learning regarding the expansionary effect of grant on system dimensioning. Normally, whoever is offering such grants aims to lower cost of supply to promote affordability, and possibly, the generation of sufficient financial returns, at least, to cover operational cost. For development partners especially, the aim is to facilitate sustainability to guarantee the ultimate developmental impacts of such investments beyond connection. But then, if grants are to result in unwanted extra cost as illustrated here, then they become counterproductive to their envisaged benefits. Thus, the second learning is that development partners or any other giver of such grants ought to be critically diligent at the design stages of mini-grids to ensure that system designers completely account for cost in their planning models or does not arbitrarily decide mini-grid sizes on the influence of “free money”.

Contrasting the constraint and unconstrained optimisation results of the operating systems, it can be seen that optimising the pure renewable systems to achieve high reliability led to increased cost owing to a surge in storage capacities especially. With the Kudorkope system for instance, pushing for 100% reliability from about 94%, results in capital cost addition of about 25% which eventually increased cost of supply. In fact, renewable integrations exceeding 80 is demonstrated to sharply drive up cost (Husein et al., 2017; Lilienthal et al., 2013) in what is described as the “hockey stick” price increase (Lilienthal et al., 2013), since additional investments in storage and control technologies become required (Lilienthal et al., 2013; Shaahid et al., 2007). This is precisely why the DG backed-up systems performed better financially than the pure renewable systems (Tables 6 & 12). Thus, the third learning is that aiming for high reliability requires diesel support to manage cost, otherwise, a small proportion of capacity shortage should become acceptable to help avoid a “hockey stick” cost effect.

The fourth learning can be drawn from the fact that over dimensioning does not guarantee reliability in the long-run but can lead to altering operating plans which if not monitored, can result in damaged equipment from overload. This will especially be the case for mini-grids without smart features as incorporated into the plants analysed here. The continuous disconnection of the inverter systems anytime the low voltage disconnection threshold is reached helped to protect the batteries from being damaged prematurely from deep discharge. However, what would have been most appropriate, is going for automation of the systems which will enable utilization of the complete smart functionalities of the inverters. Herein, rather than disconnecting, the inverters would have signaled the gensets to start on reaching the low voltage rate and will balance serving the loads with charging the batteries. This will improve capacity utilization of the gensets, maintain DoD which can extend battery life, and avoid inappropriate interference by local operators who were found to lack the required technical expertise. Above all, customer confidence can be maintained.

As observed from the financial analysis, mini-grids can impose significant financial burden on governments and utilities. On average, the annual revenue requirement comes to about \$9200 annually when initial investments are funded by grant, and about \$34,500 without grant. The often talked about donor fatigue points to the fact that grants will not always be available and most mini-grid projects will need to be funded from loans contracted by governments. Even in Ghana where a relatively small number (300) (Energy Commission of Ghana, 2019) of mini-grids are planned to be deployed by 2030, the financial burden on government and the utilities involved could amount to \$10.4 million annually. With a debt distressed economy¹ (IMF, 2020) and a struggling power sector² (Ministry of Energy, 2019) as it is for many utilities in

Sub-Saharan Africa (World Bank, 2016), it may be difficult to meet such obligations, which may affect sustainable operations of the mini-grid plants, if they are managed to be built. Thus, the fifth learning is that cost-cutting measures (including building to fit, demand side management and positioning mini-grid where they are economical than the grid) must be an integral part of sustainable mini-grid development programmes. This is especially crucial because one of the valued contributions of mini-grids to electrification programs (as expected of the ones analysed here) is that they represent a cheaper option for areas too far from the grid or with difficult terrains (SEforALL, n.d.; ARE, 2014), otherwise, conventional grid extension is generally more economical than mini-grids.

Linked to the preceding is the issue of tariffs which should seek to balance affordability with cost recovery. The usual proposal is that such tariffs must not exceed pre-electrification expense on alternative energy (kerosene, dry cell, candles, etc.), but should approach willingness to pay and stay comparable to grid tariffs for reasons of fairness. This aims to exploit affordability limits to reasonably recover cost for a reduced burden on governments, as opposed to full cost recovery. Yet, the alternative energy cost of \$0.60/kWh and the average monthly willingness to pay of \$9.57³ as reported for Kudorkope for instance (PWC, 2012), are observed to be considerably higher than the applied average tariff of 0.115/kWh and the average monthly charge of \$5.12. Accordingly, it would still have been affordable and fair to push tariffs towards the average residential grid tariff of \$0.21/kWh to narrow the observed viability gaps. Besides, just as cost of alternative energy will not remain fixed, so are tariffs not expected to be fixed, and in Ghana, an automatic tariff adjustment formula is applied in this regard. As demonstrated by the financial results, pushing tariffs towards willingness to pay and alternative energy cost limits, which in the least will come to grid parity level here, and applying PURC's periodic tariff reviews, enables an average recovery of 90 cents for every dollar invested (i.e. Table 12). Hence, the sixth and final learning is that, efficiently designed tariffs combined with appropriate regulation, can narrow the burden of mini-grid revenue shortfalls on governments and utilities, and will most importantly, guarantee the sustainability of mini-grids into the future.

Conclusion

This study adopted an integrated scenario modelling approach to investigate design accuracy, power supply challenges and financial sustainability of two existing mini-grids in Ghana. It also examined the economic competitiveness of grid extension relative to the mini-grid plants. As shown by the results, both mini-grids were determined to be oversized and should ideally be able to meet the existing loads, or even the design loads without any issues. The challenges experienced with power supply were found to be linked to the operating decision of running the plants as pure renewable energy systems which is worsened by the fact that the batteries under both mini-grids have now reached their terminal points.

The extent of capacity constraints making it difficult for the plants to operate reliably were found to be 8% in Kudorkope and 14% in Peditorkope. This stems from altering operations of the Kudorkope plant from a PV-DG-Battery system to a PV-Battery system. And in varying the Peditorkope plant from a PV-WT-DG-Battery system to a PV-WT-Battery system. Two reasons were determined to account for the over dimensioned nature of the plants: initial load over-estimation and the influence of grant. Beyond being oversized, the mini-grid plant in Peditorkope was also found to be inefficiently configured given the site resources. Cost of electricity supply was found to be significantly higher than the current tariffs being charged in all cases making it impossible for the recovery of even O&M cost at the existing loads. Thus, the attainment of financial sustainability appears a nonstarter.

³ Willingness to pay and alternative energy expense are converted using 2016 exchange rate to enable comparison.

¹ Ghana's debt to GDP ratio stood at 76.7% with an overall fiscal balance of –16.4% as of October 2020.

² Power sector revenue shortfalls stood at \$2.7 billion in 2018 and is projected to reach \$12.5 billion by 2023.

However, pushing tariffs towards grid parity levels and applying appropriate tariff regimes or regulations similar to what applies to centralised grid customers hold the potential to significantly reduce the burden of unmet revenues on government and the utilities involved, which will most importantly guarantee sustainability of the mini-grid plants into the future. The mini-grids analysed were also observed to be cost-ineffective compared to grid extension, which signals inefficiency in the electrification planning process. We conclude from the totality of the analysis that the design, operations and siting of future mini-grid plants can be improved. In this regard, geospatial planning tools can help with more appropriate siting decisions while simulation and optimisation tools like HOMER can play the fundamental role of ensuring efficient allocation of, or use of resources allocated to generation sizing as demonstrated in this paper. Guaranteeing the efficacy of the preceding requires that all input data, especially the load input, must be generated with careful diligence. And from the tariff rate setting perspective, the analysis in this paper show that carefully exploiting willingness to pay, and pre-electrification alternative energy expense can result in tariffs that remain fair and affordable but competitive.

While the study attempts to cover as many issues as possible with the scenario modelling, it is just impossible to exhaust all the relevant issues in a single paper such as this. To that extent, the implications of this study could benefit from further scenarios. For instance, it would have been good to compare the performance of lithium-ion batteries in the scenarios with the lead acid batteries used (and even more). This may generate useful implications in choosing between batteries for future mini-grids deployment. Also, the paper's usefulness could be expanded by conducting further analysis with adjusted versions of the load profiles. The object will be to explore how the community loads can be restructured to ensure efficient use of the mini-grid plants as well as inform the curation of load profiles for future mini-grid designs. These issues, can inter alia be explored by future similar studies. But without doubt, the insights offered by the current analysis, will be useful to policy makers and the mini-grid implementing agencies in Ghana, with broader usefulness for all stakeholders interested in the sustainable deployment of mini-grids.

Declaration of competing interest

The authors wish to confirm that there are no known conflicts of interest associated with this publication, and there has been no significant financial support for this work that could have influenced its outcome. The authors also declare as follows;

That the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

That we have given due consideration to the protection of intellectual property associated with this work and that there are no impediments to publication, including the timing of publication, with respect to intellectual property. In so doing, it is confirmed that we have followed the regulations of our institutions concerning intellectual property.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.esd.2022.04.009>.

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